Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 4

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 3

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : -0.000000
S   : -0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.629656
<D> / site : 0.271609

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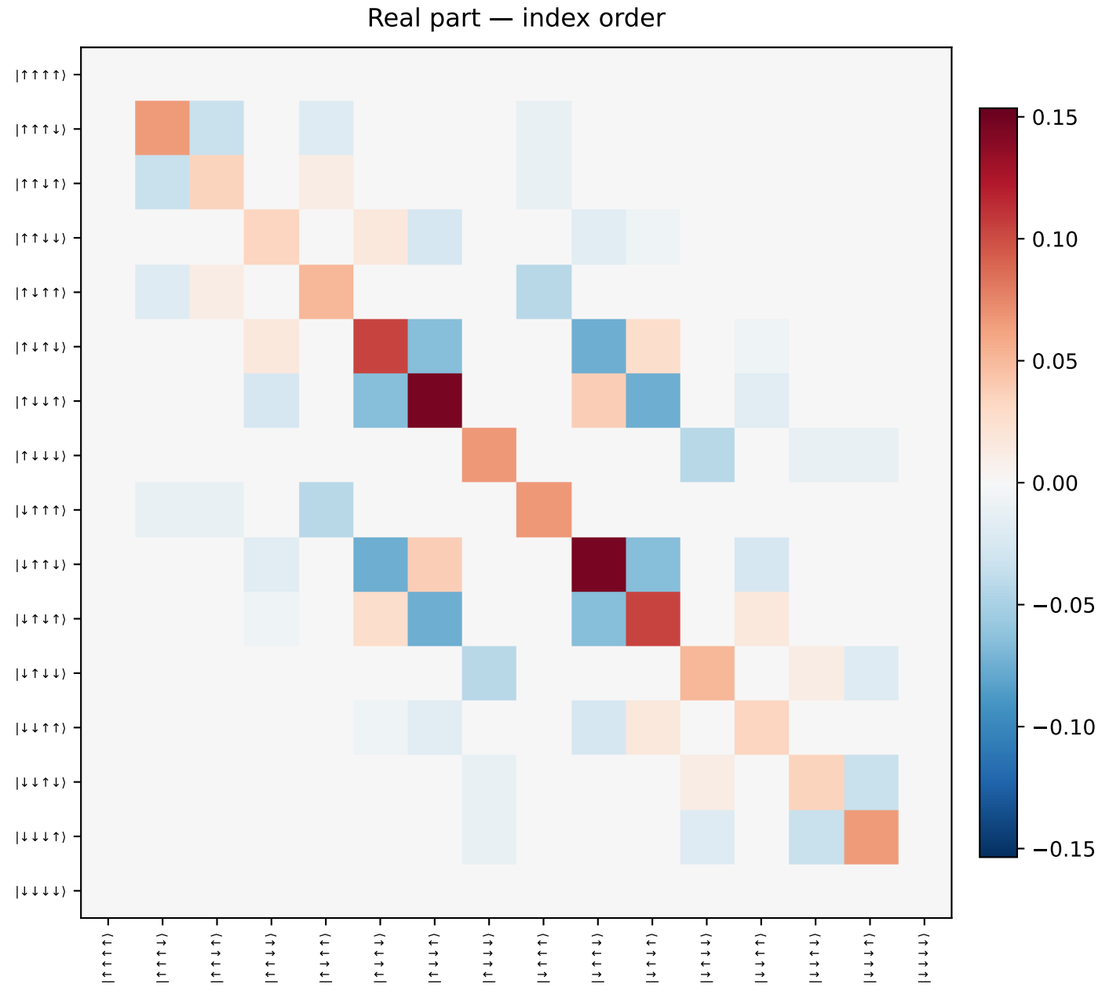
Body order : 4
Clusters   : 15

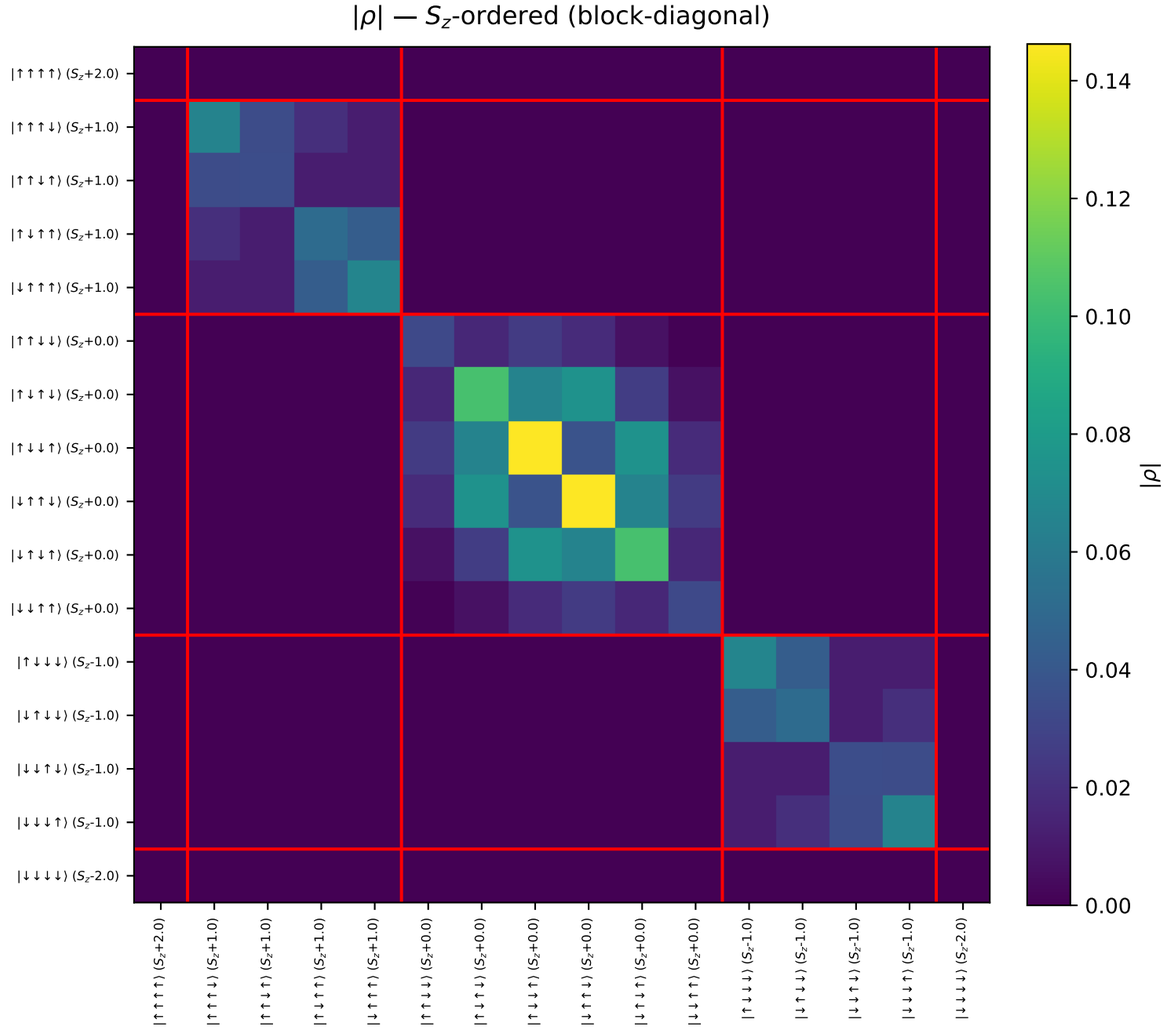
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Summary — 4-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2,3	0.155426	2.099764	—	0.000000	0.055258	0.522026	STRONGLY MIXED
0,1,2,4	0.158475	1.997490	—	0.000000	0.035902	0.431177	STRONGLY MIXED
0,1,2,5	0.229065	1.762692	—	0.000000	0.042761	0.461830	MIXED
0,1,3,4	0.148104	2.023123	—	0.000000	0.049094	0.383196	STRONGLY MIXED
0,1,3,5	0.180530	1.893314	—	0.000000	0.022949	0.294033	STRONGLY MIXED
0,1,4,5	0.342104	1.572341	—	0.077728	0.081116	0.445909	MIXED
0,2,3,4	0.149024	2.049578	—	0.000000	0.007271	0.379896	STRONGLY MIXED
0,2,3,5	0.137008	2.142720	—	0.000000	0.058832	0.381587	STRONGLY MIXED
0,2,4,5	0.121609	2.205649	—	0.000000	0.049634	0.415629	STRONGLY MIXED
0,3,4,5	0.129993	2.131813	—	0.000000	0.022378	0.295047	STRONGLY MIXED
1,2,3,4	0.158254	2.042876	—	0.000000	0.048435	0.411075	STRONGLY MIXED
1,2,3,5	0.300189	1.582016	—	0.000000	0.020941	0.344744	MIXED
1,2,4,5	0.179918	1.912561	—	0.000000	0.039429	0.242406	STRONGLY MIXED
1,3,4,5	0.181960	1.911215	—	0.000000	0.058349	0.245925	STRONGLY MIXED
2,3,4,5	0.164352	2.016905	—	0.000000	0.024065	0.338947	STRONGLY MIXED

4-Body RDM (Sites 0,1,2,3)

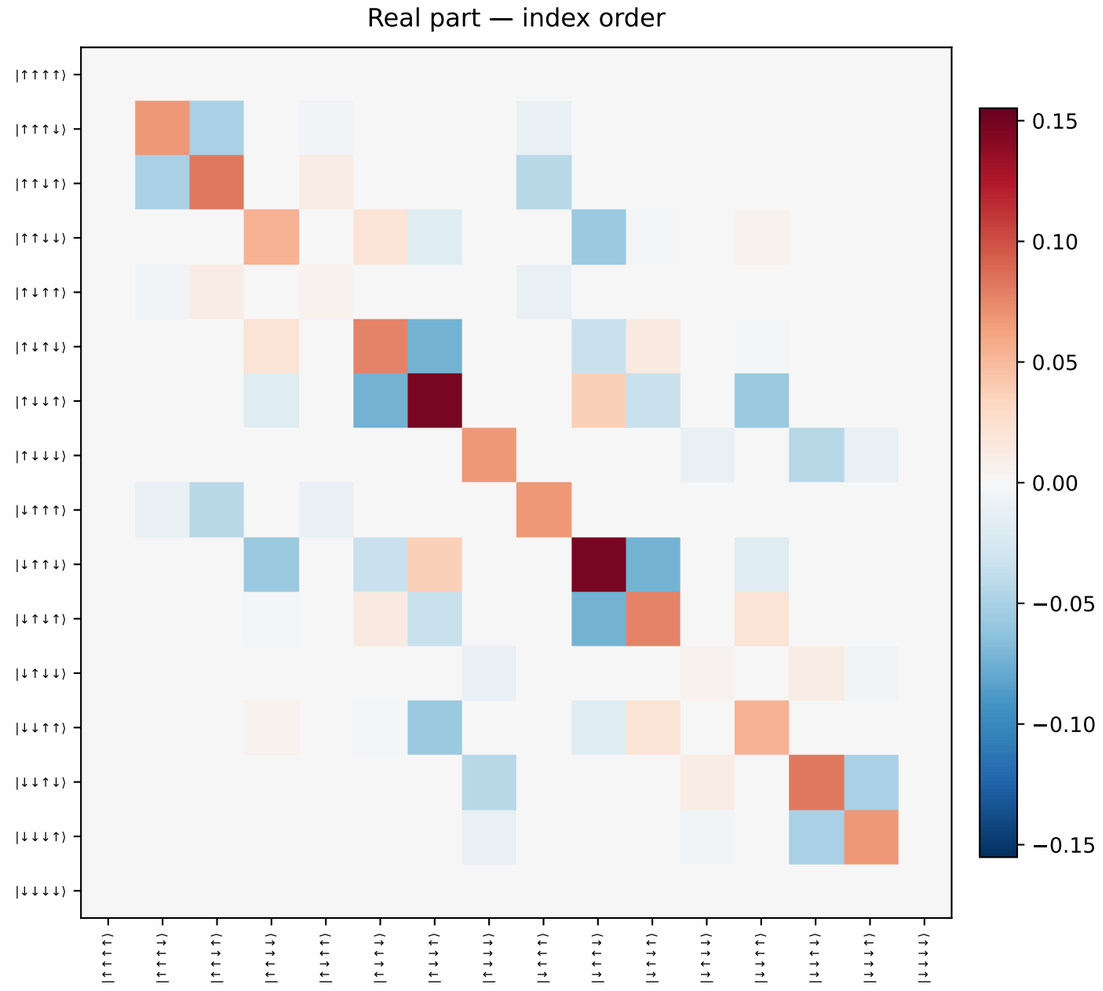


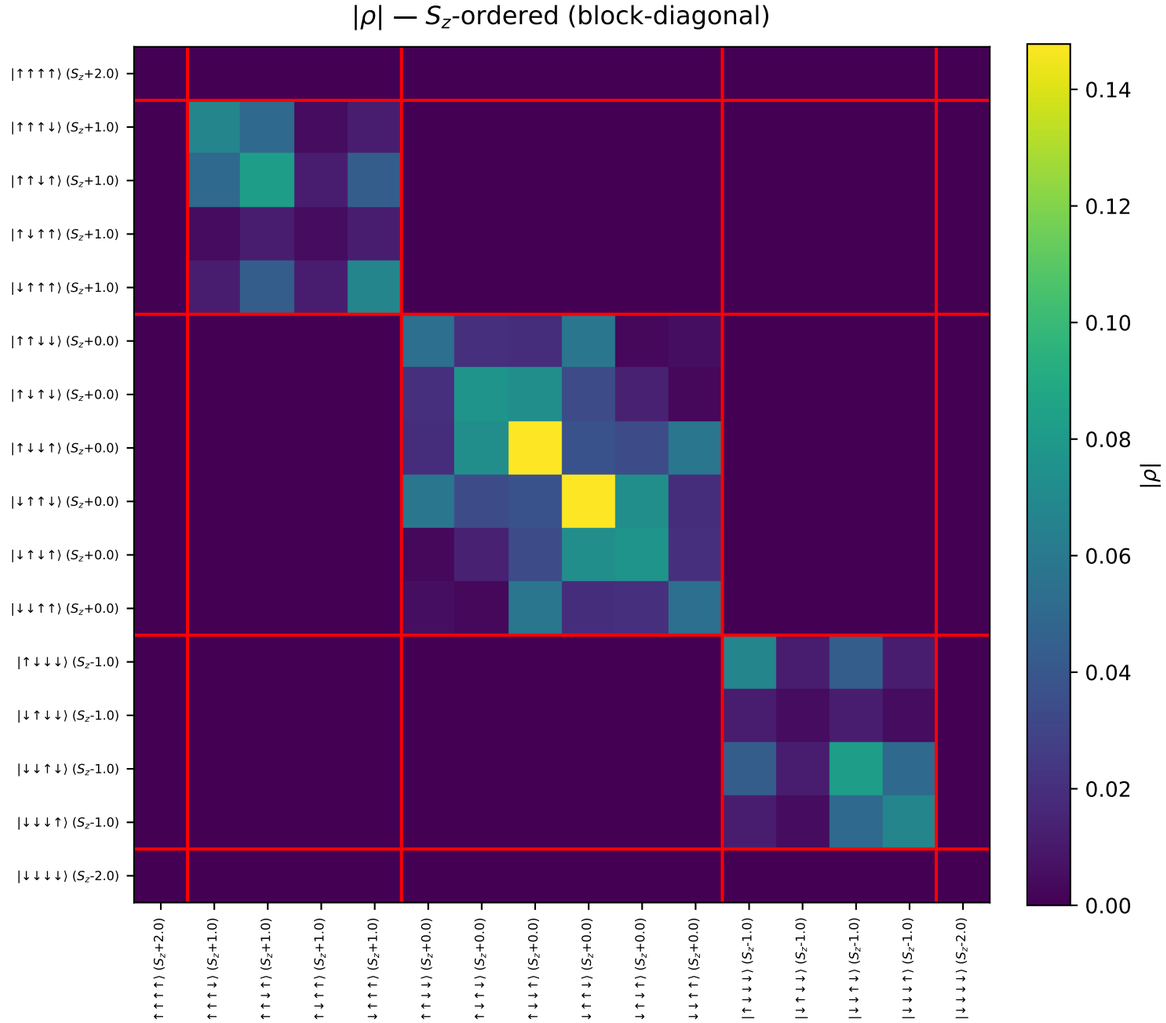


Trace=1.000000 Purity=0.155426 Entropy=2.099764 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.055258 ≤ C_{UB} = 0.522026 verdict: GME

4-Body RDM (Sites 0,1,2,4)

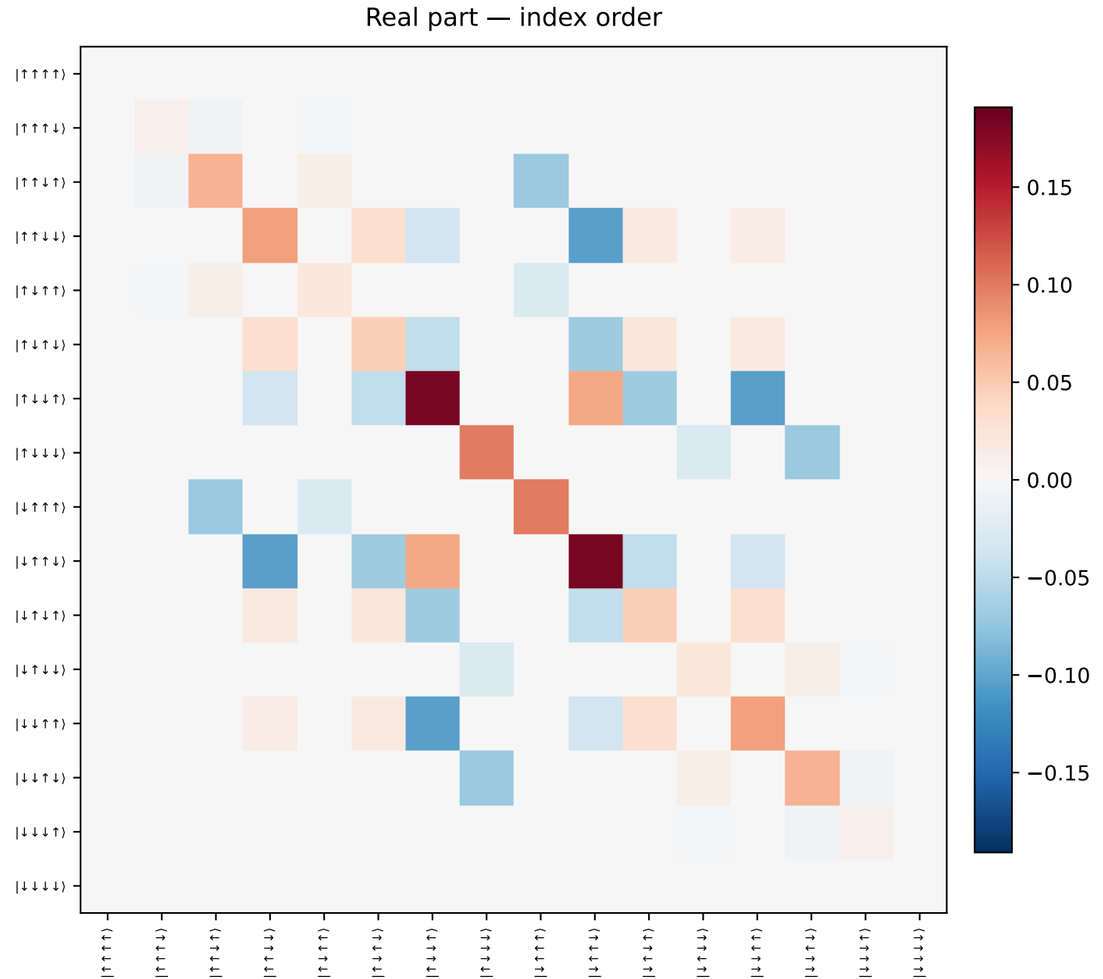


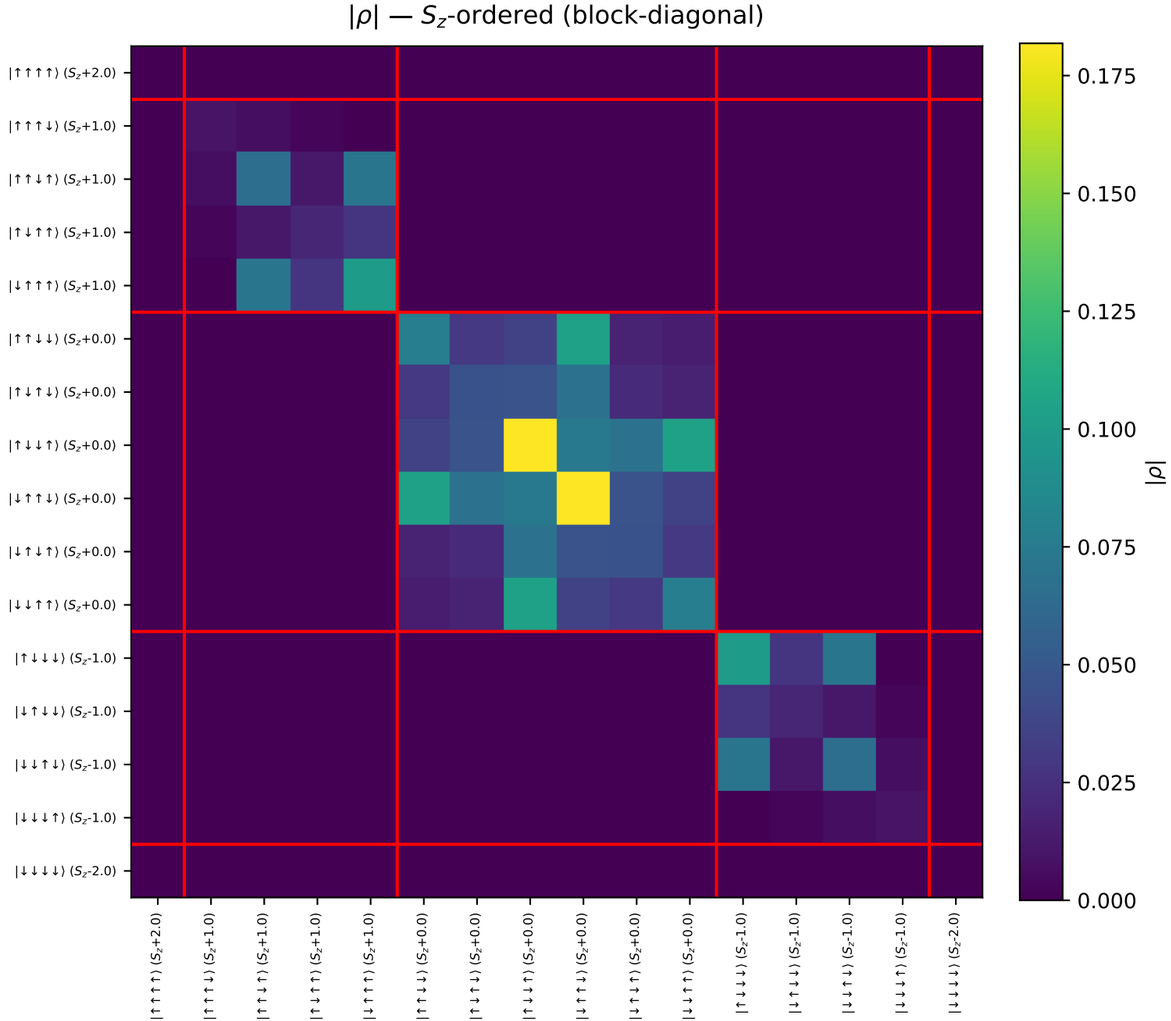


Trace=1.000000 Purity=0.158475 Entropy=1.997490 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.035902 ≤ C_{UB} = 0.431177 verdict: GME

4-Body RDM (Sites 0,1,2,5)

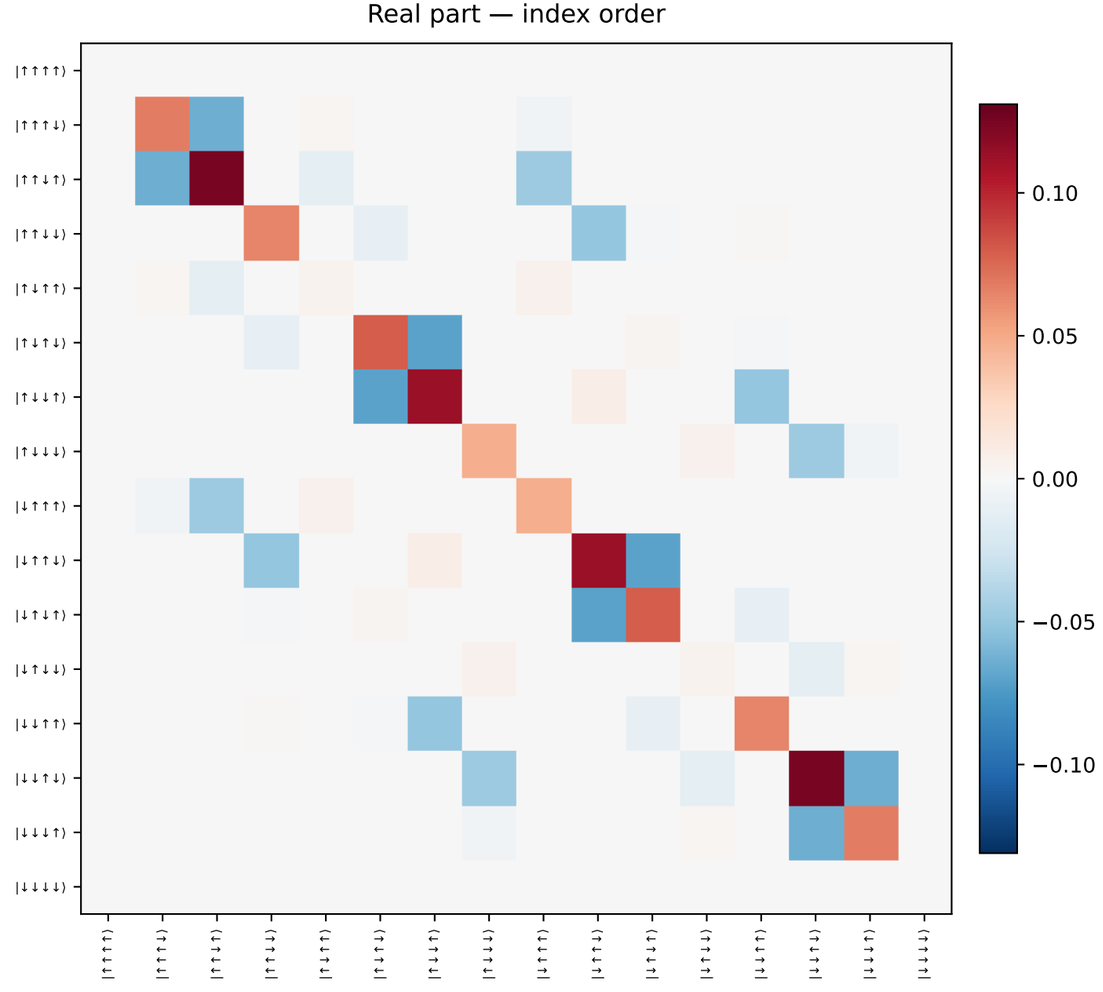


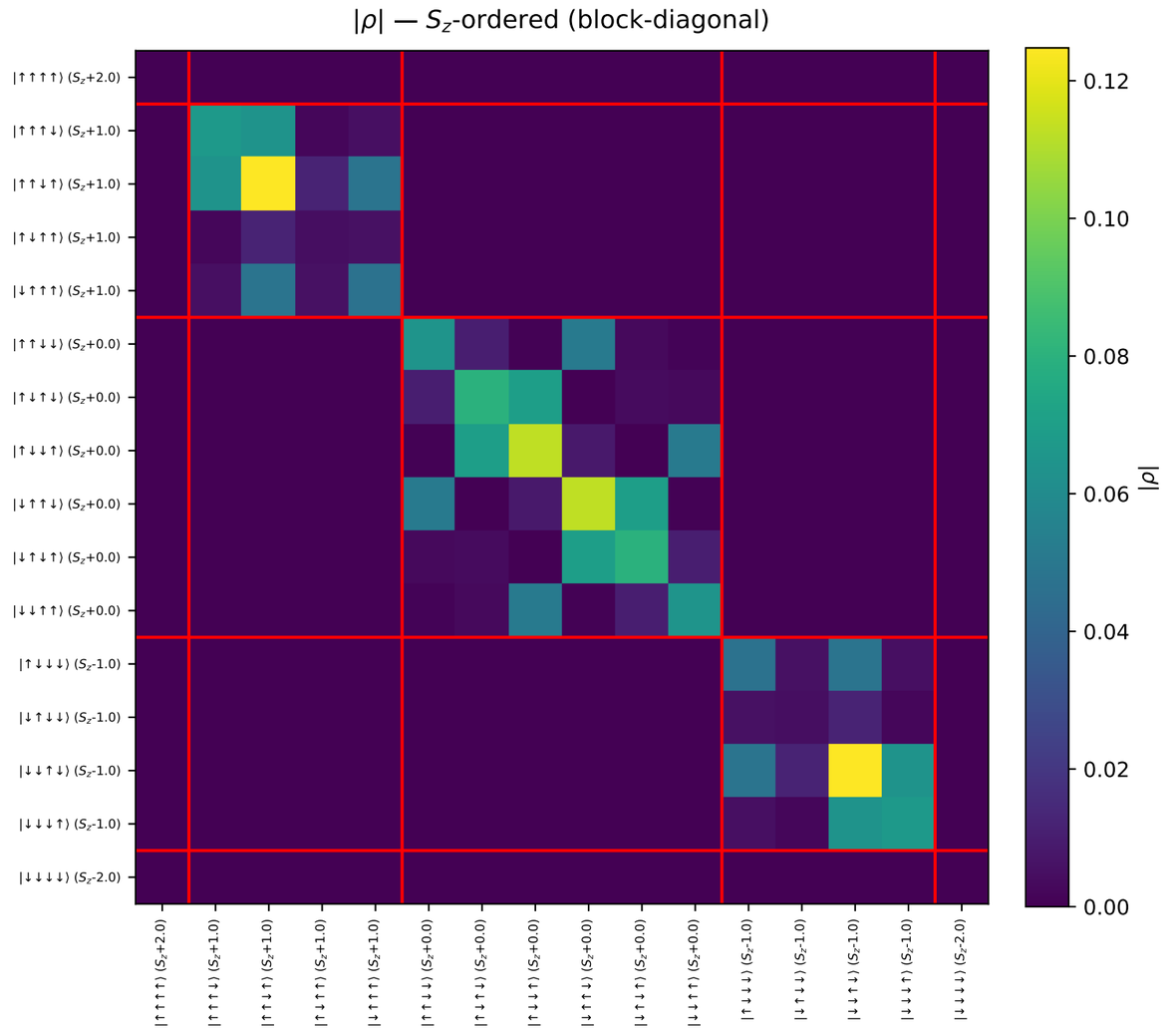


Trace=1.000000 Purity=0.229065 Entropy=1.762692 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.042761 ≤ C_{UB} = 0.461830 verdict: GME

4-Body RDM (Sites 0,1,3,4)

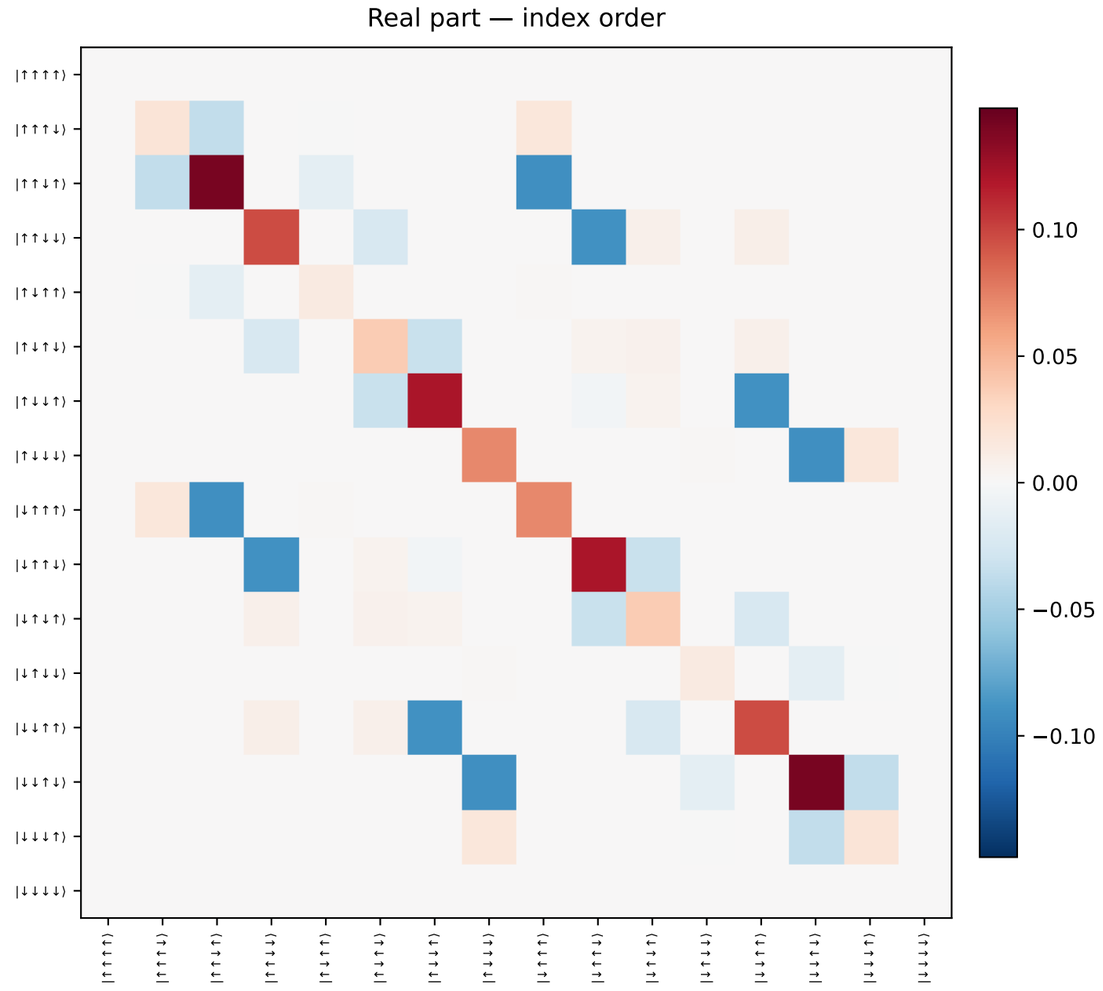


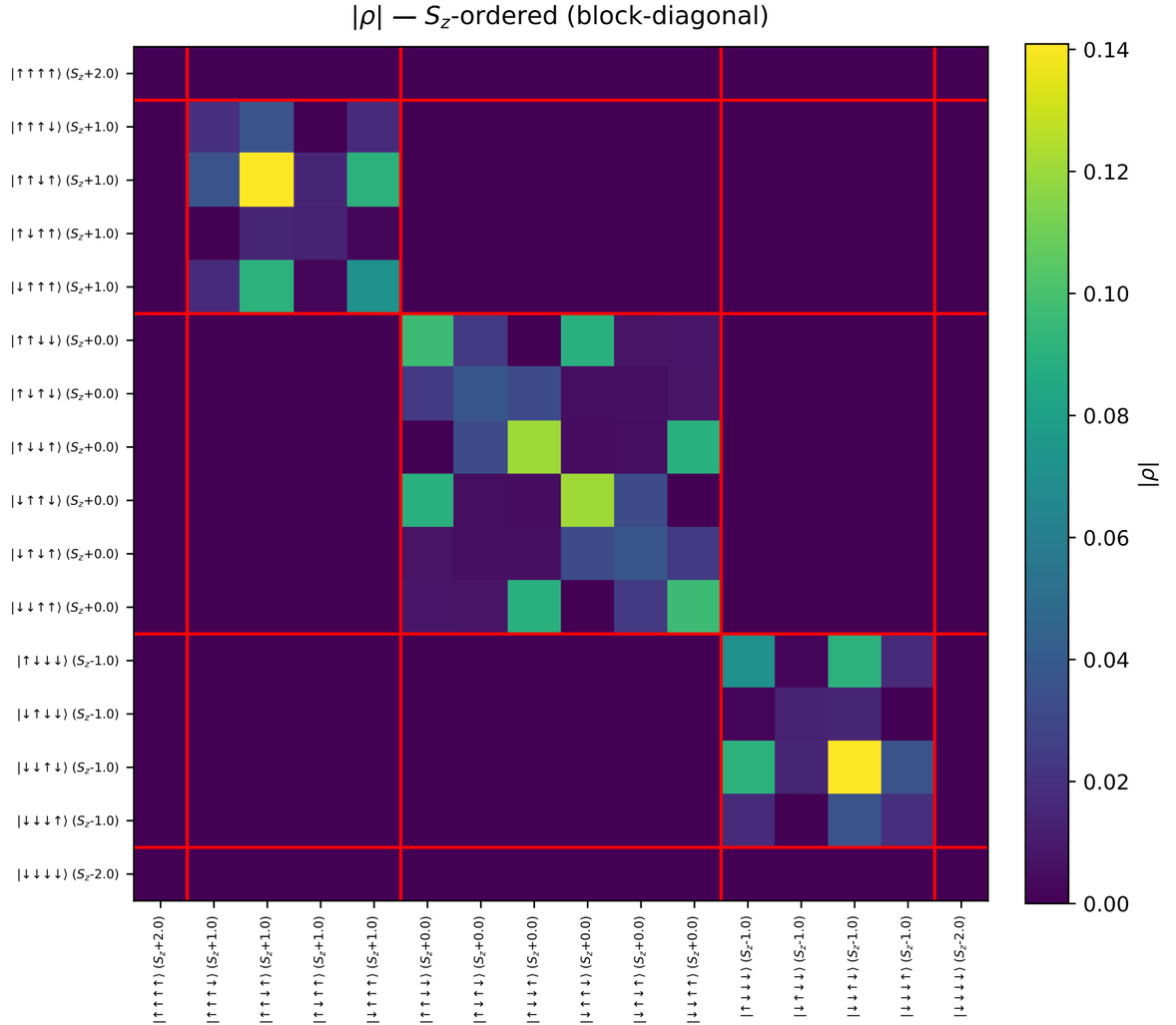


Trace=1.000000 Purity=0.148104 Entropy=2.023123 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.049094 ≤ C_{UB} = 0.383196 verdict: GME

4-Body RDM (Sites 0,1,3,5)

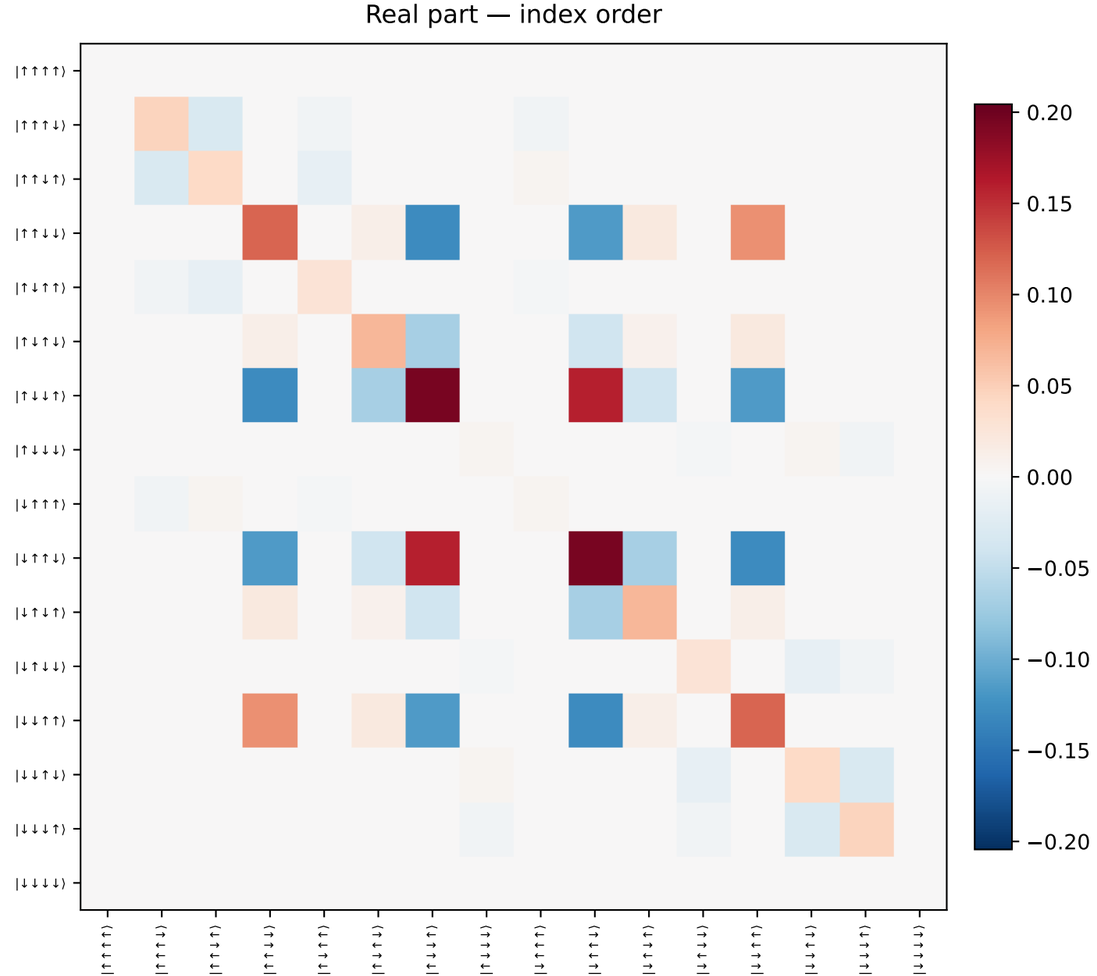


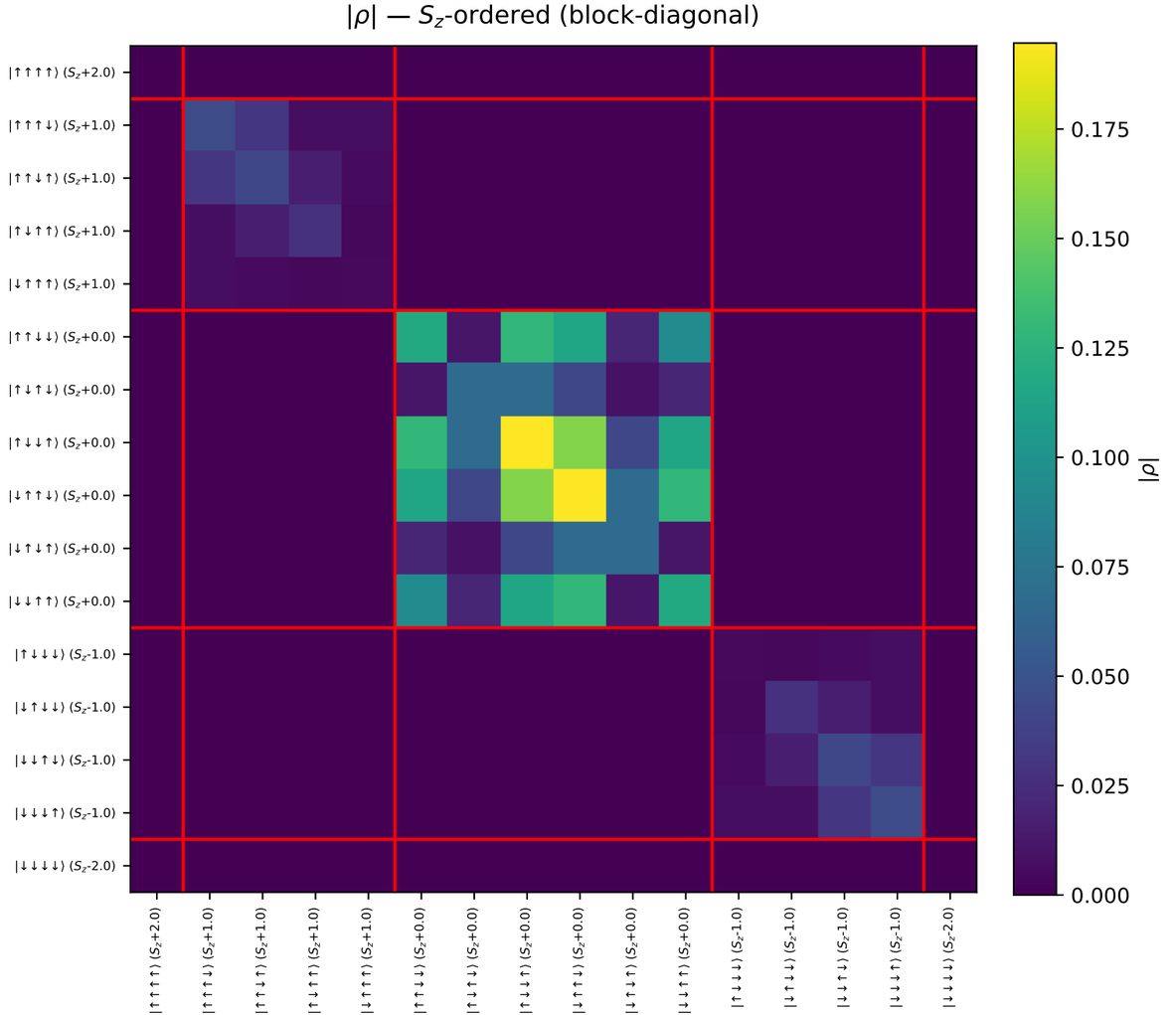


Trace=1.000000 Purity=0.180530 Entropy=1.893314 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.022949 ≤ C_{UB} = 0.294033 verdict: GME

4-Body RDM (Sites 0,1,4,5)

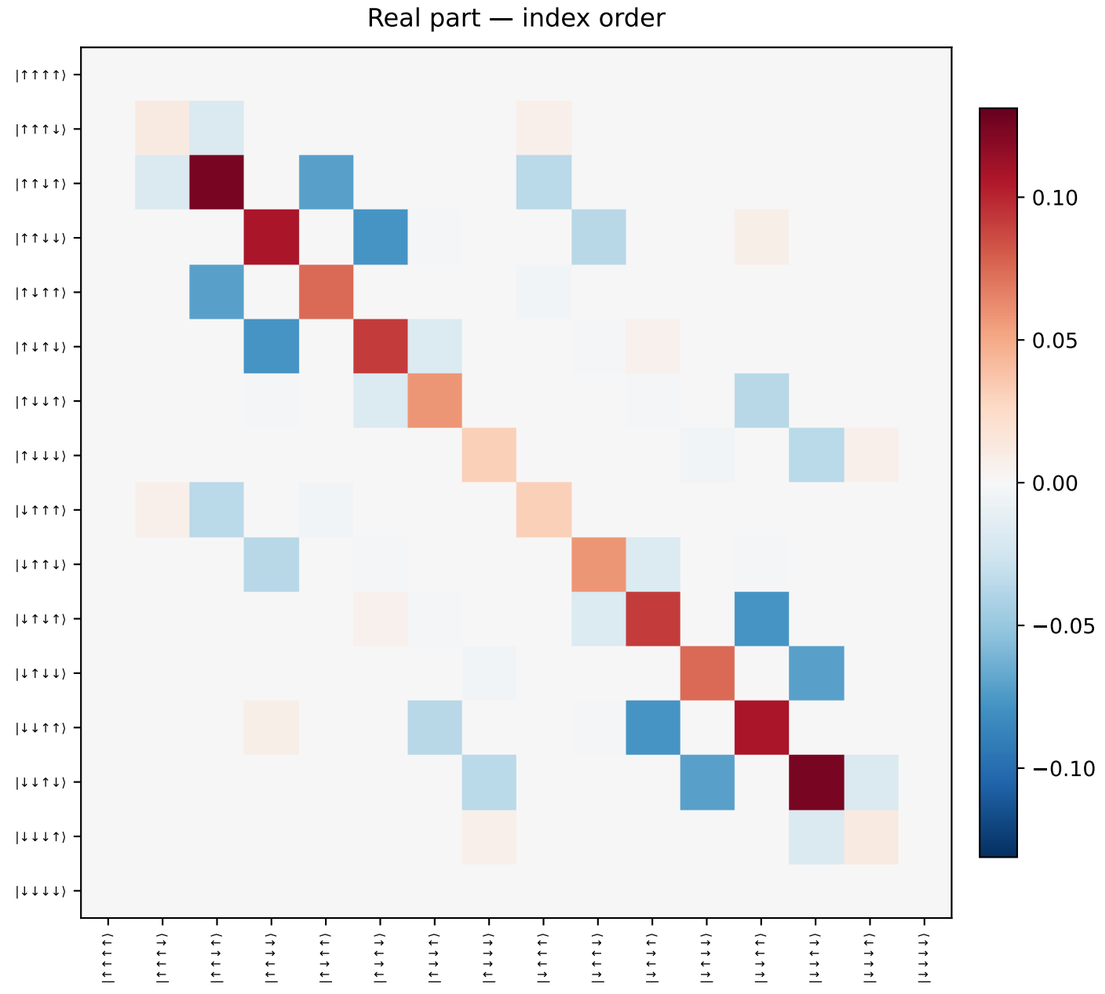


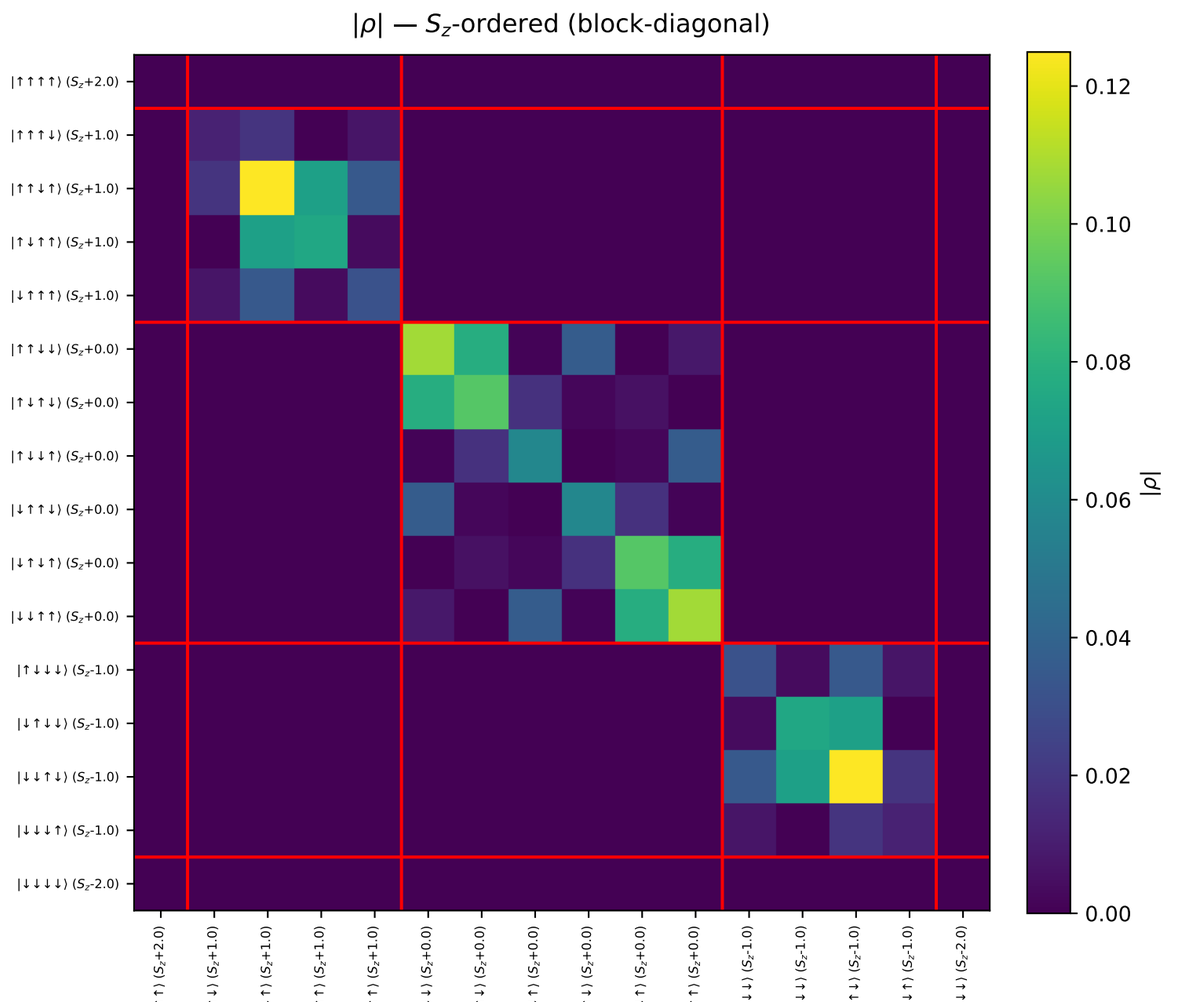


Trace=1.000000 Purity=0.342104 Entropy=1.572341 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.077728 ≤ C_{GME} = 0.081116 ≤ C_{UB} = 0.445909 verdict: GME

4-Body RDM (Sites 0,2,3,4)

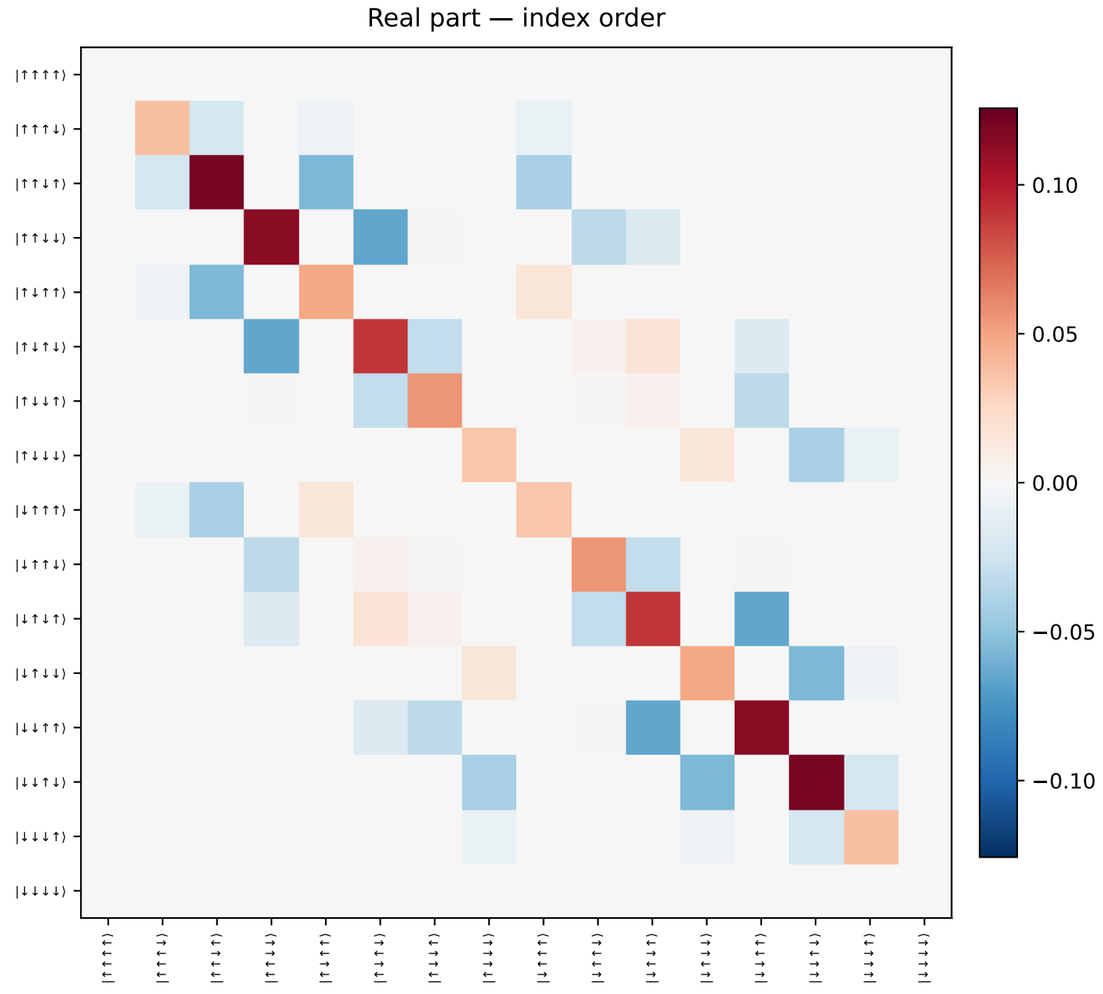


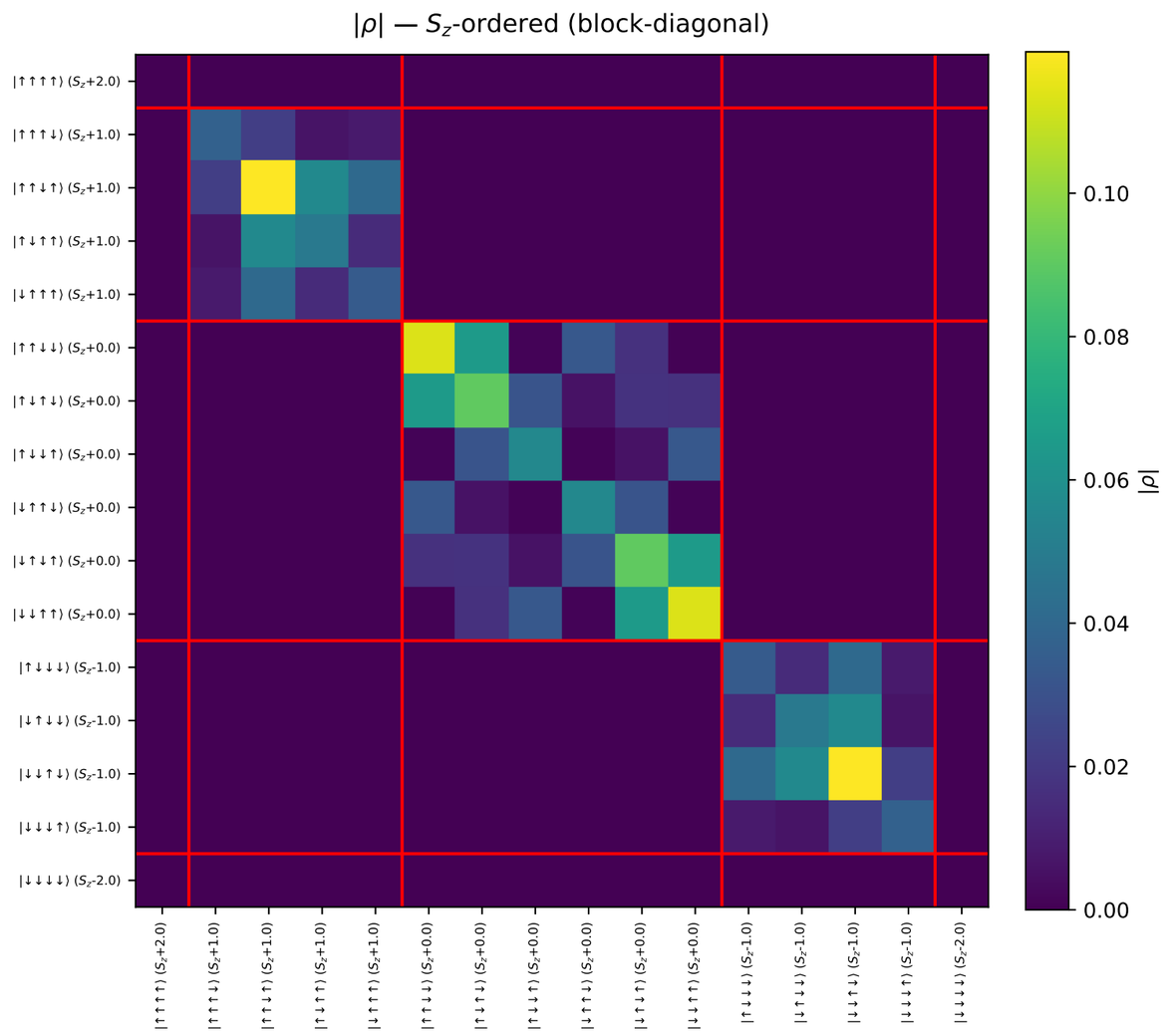


Trace=1.000000 Purity=0.149024 Entropy=2.049578 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.007271 ≤ C_{UB} = 0.379896 verdict: GME

4-Body RDM (Sites 0,2,3,5)

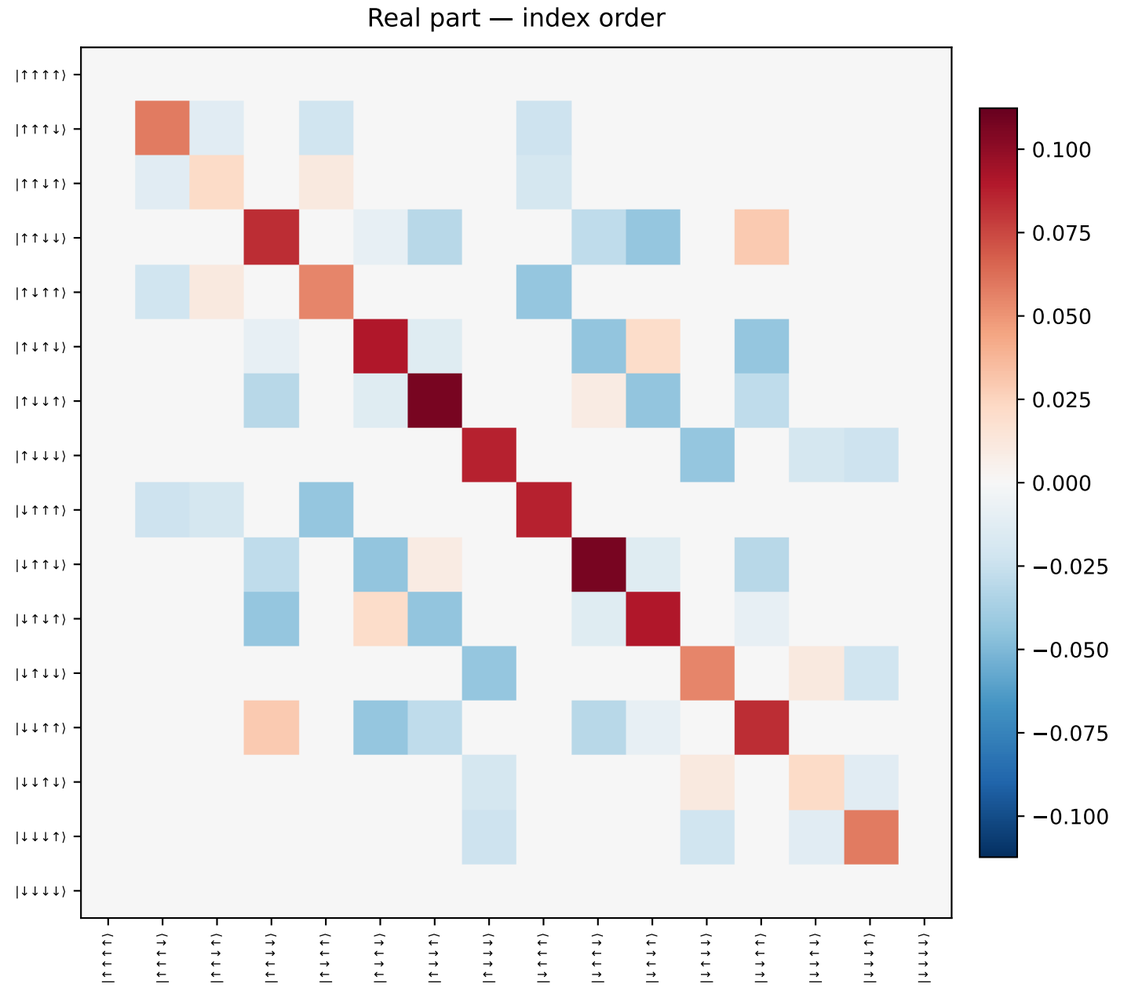


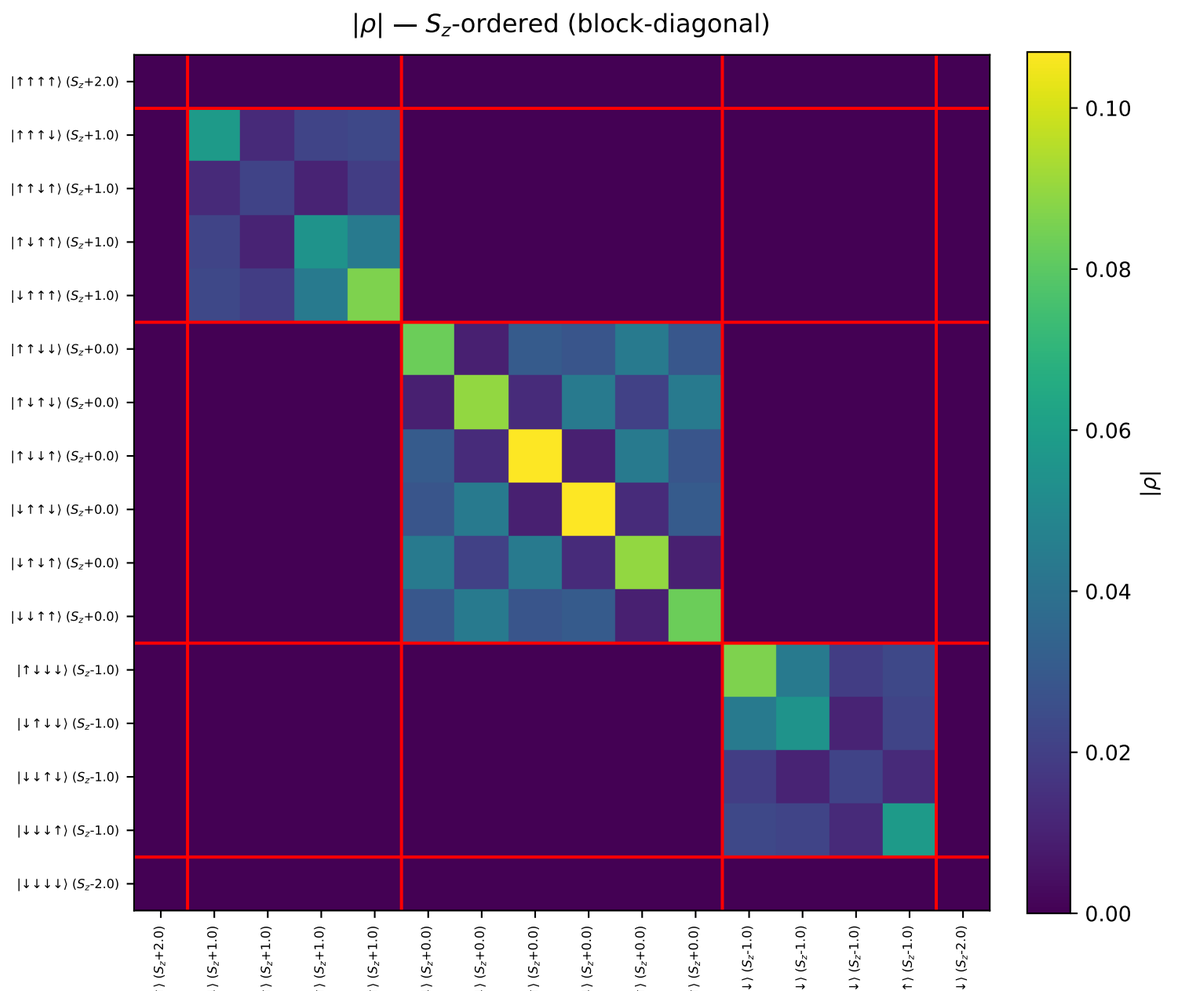


Trace=1.000000 Purity=0.137008 Entropy=2.142720 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.058832 ≤ C_{UB} = 0.381587 verdict: GME

4-Body RDM (Sites 0,2,4,5)

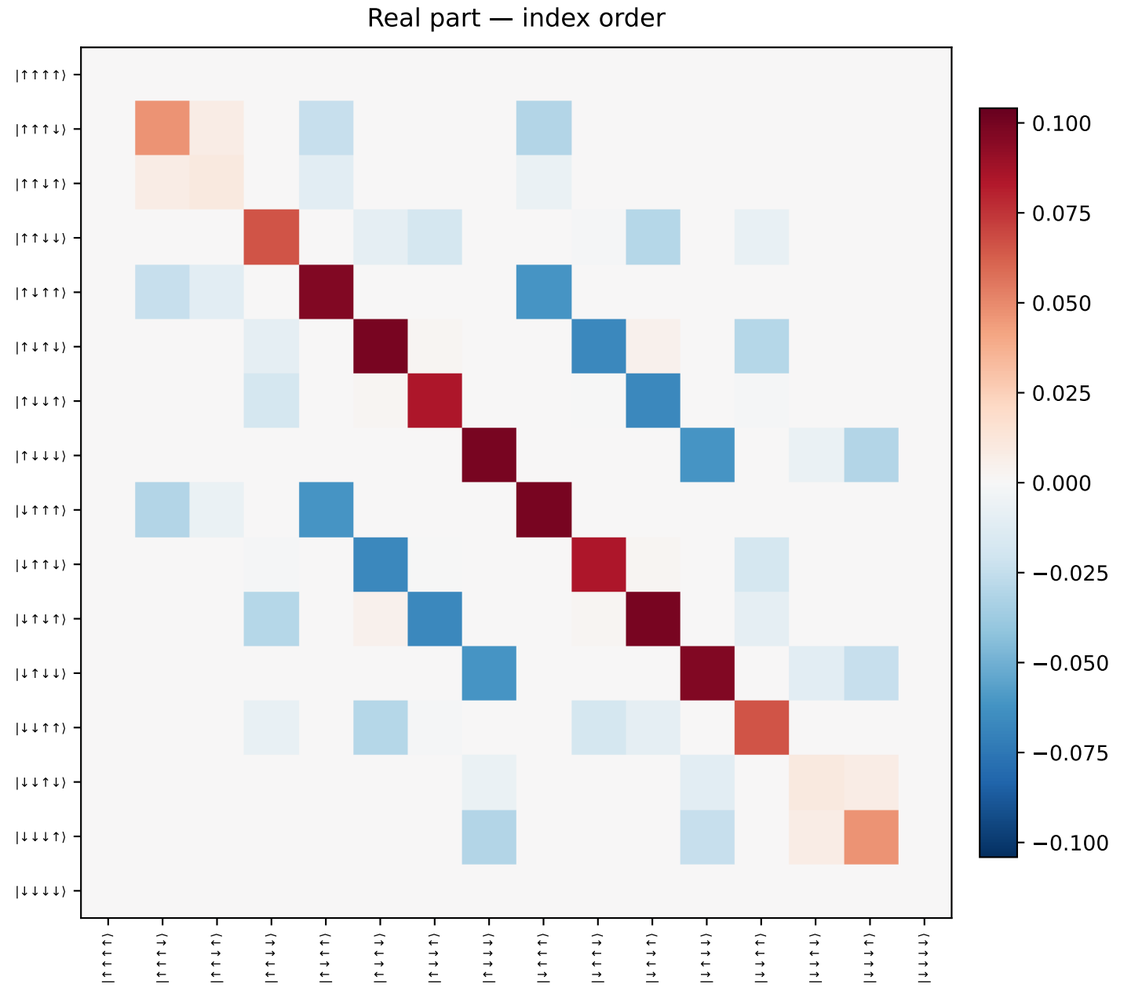


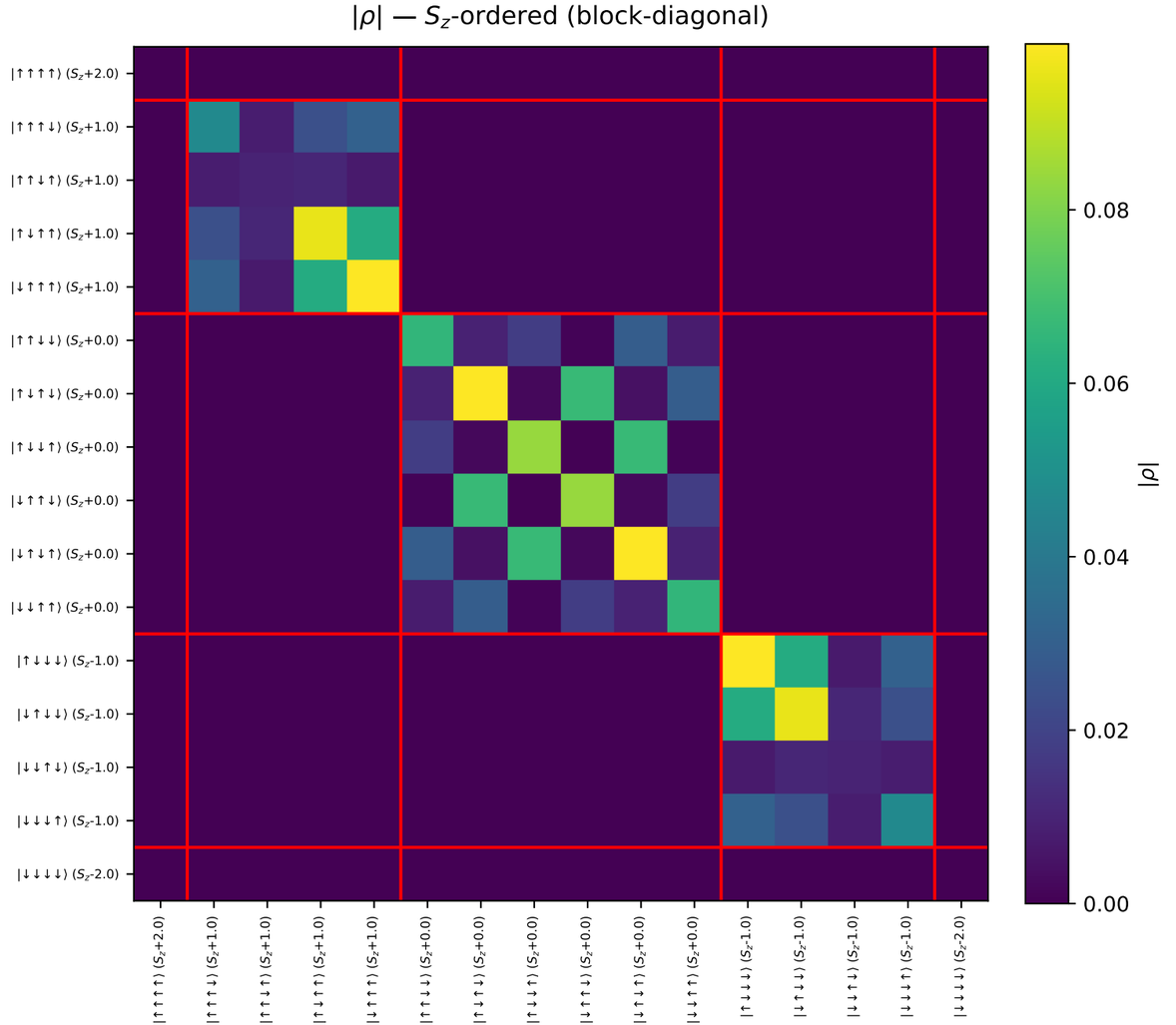


Trace=1.000000 Purity=0.121609 Entropy=2.205649 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.049634 ≤ C_{UB} = 0.415629 verdict: GME

4-Body RDM (Sites 0,3,4,5)

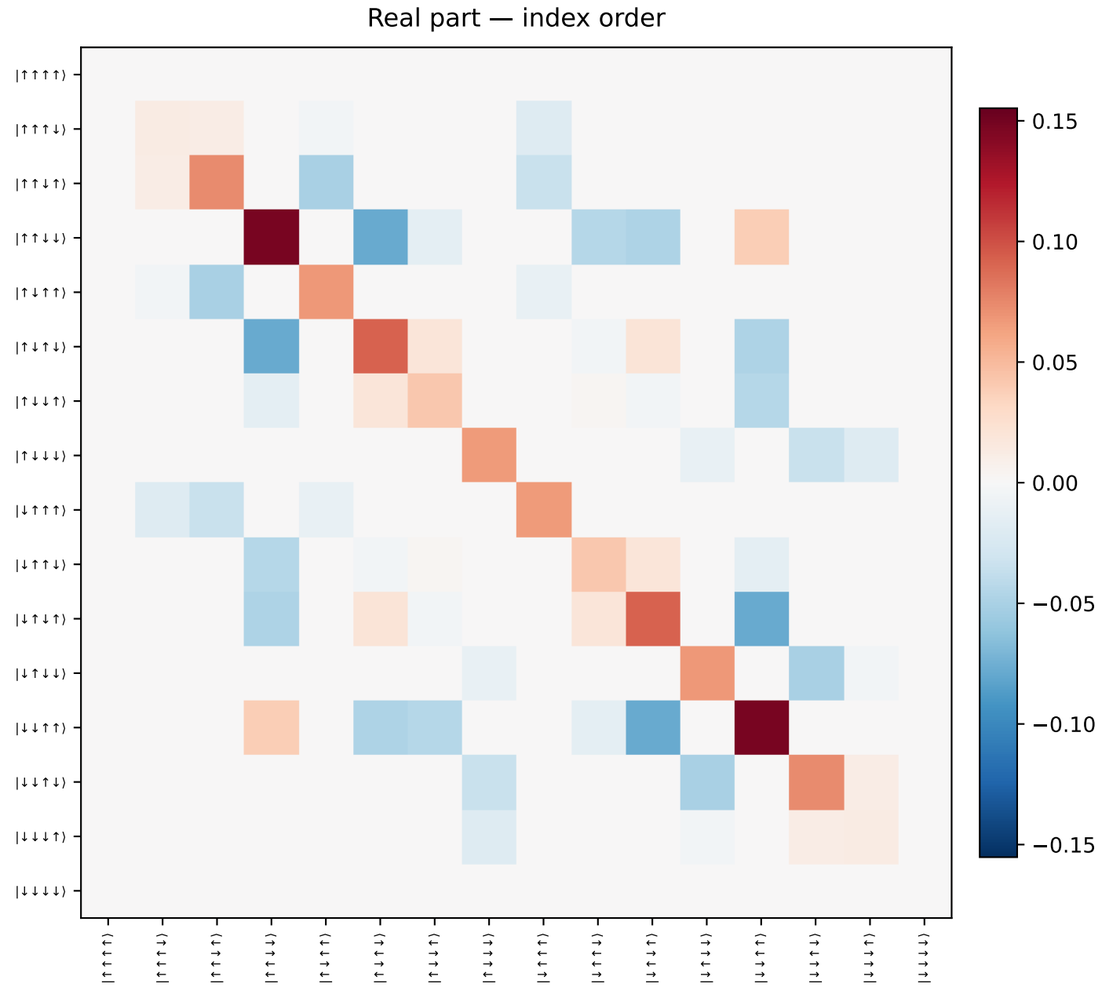


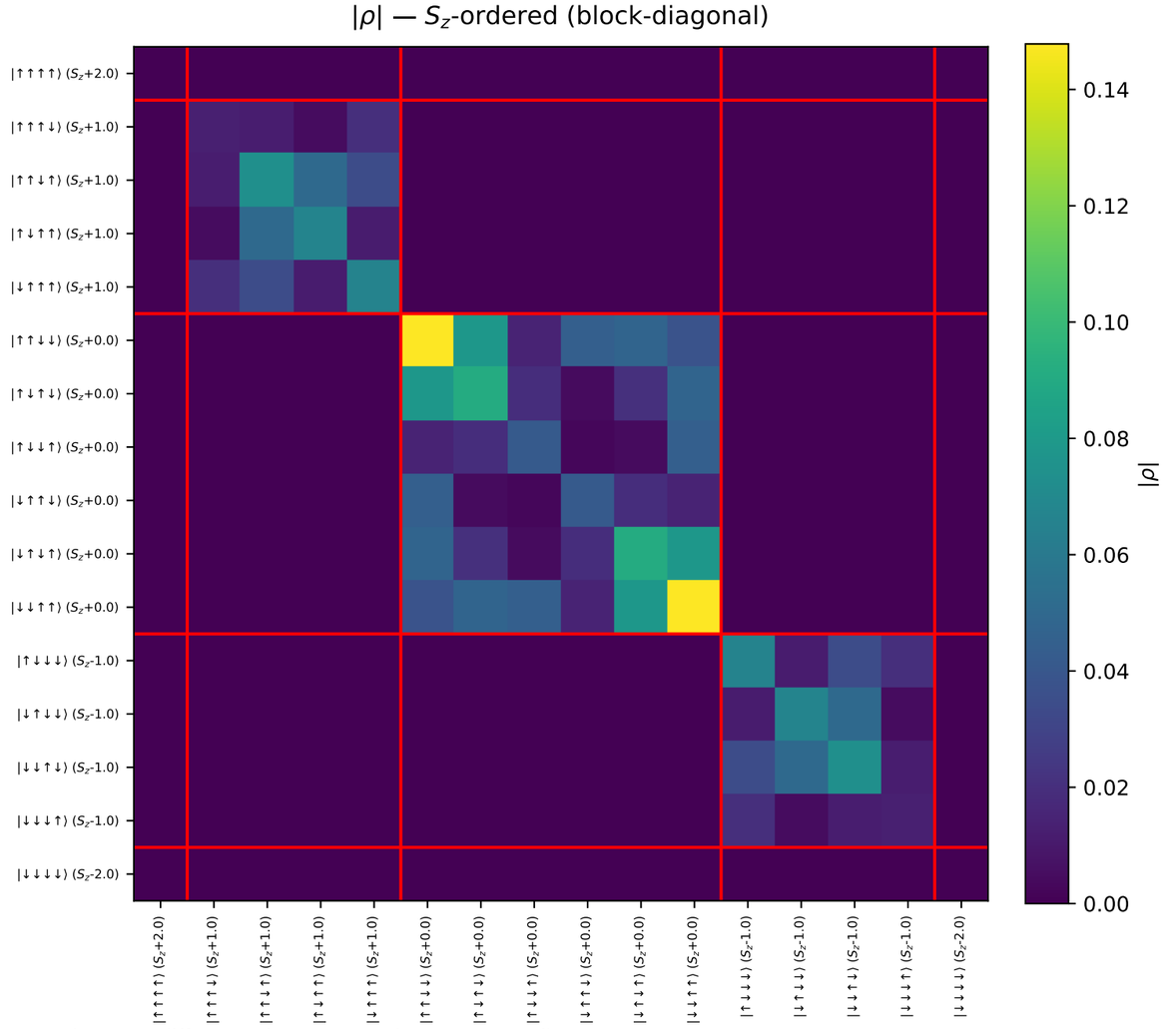


Trace=1.000000 Purity=0.129993 Entropy=2.131813 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.022378 ≤ C_{UB} = 0.295047 verdict: GME

4-Body RDM (Sites 1,2,3,4)

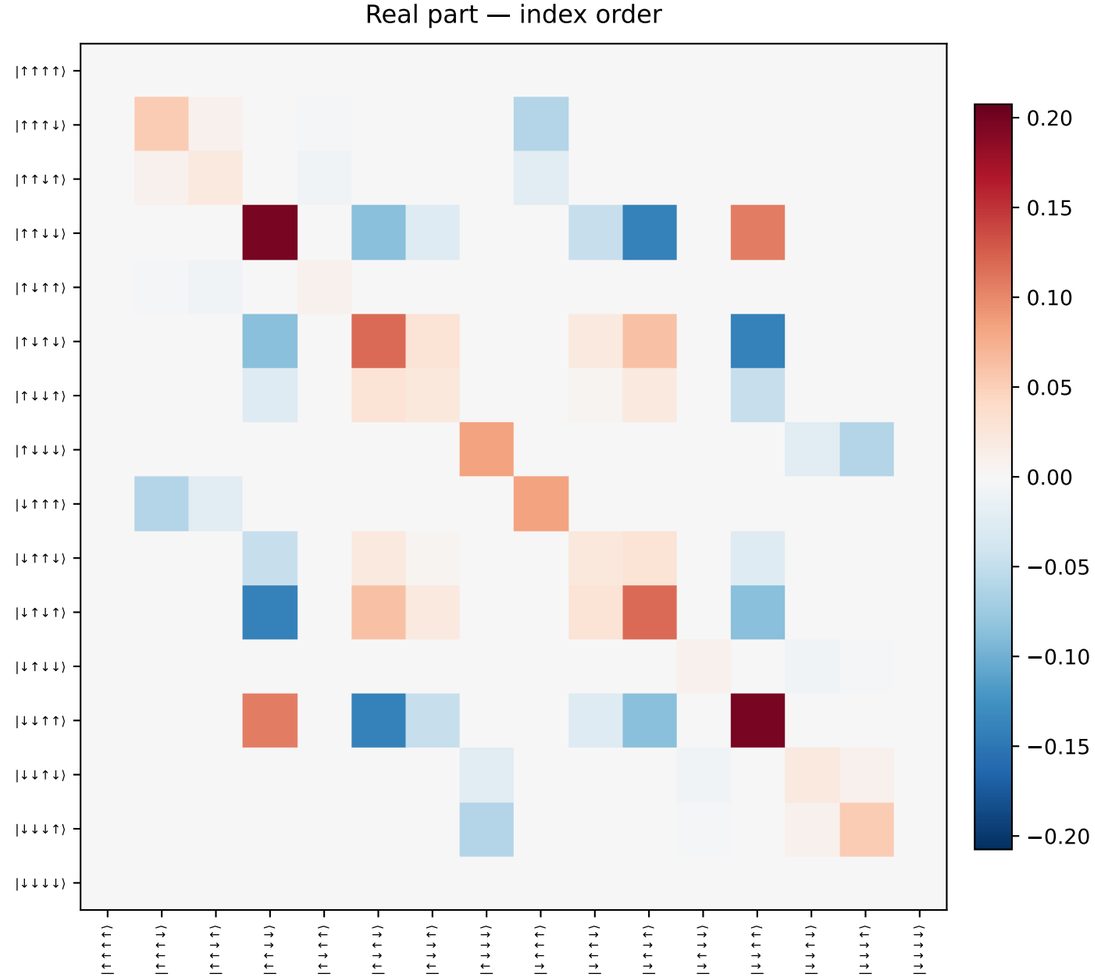


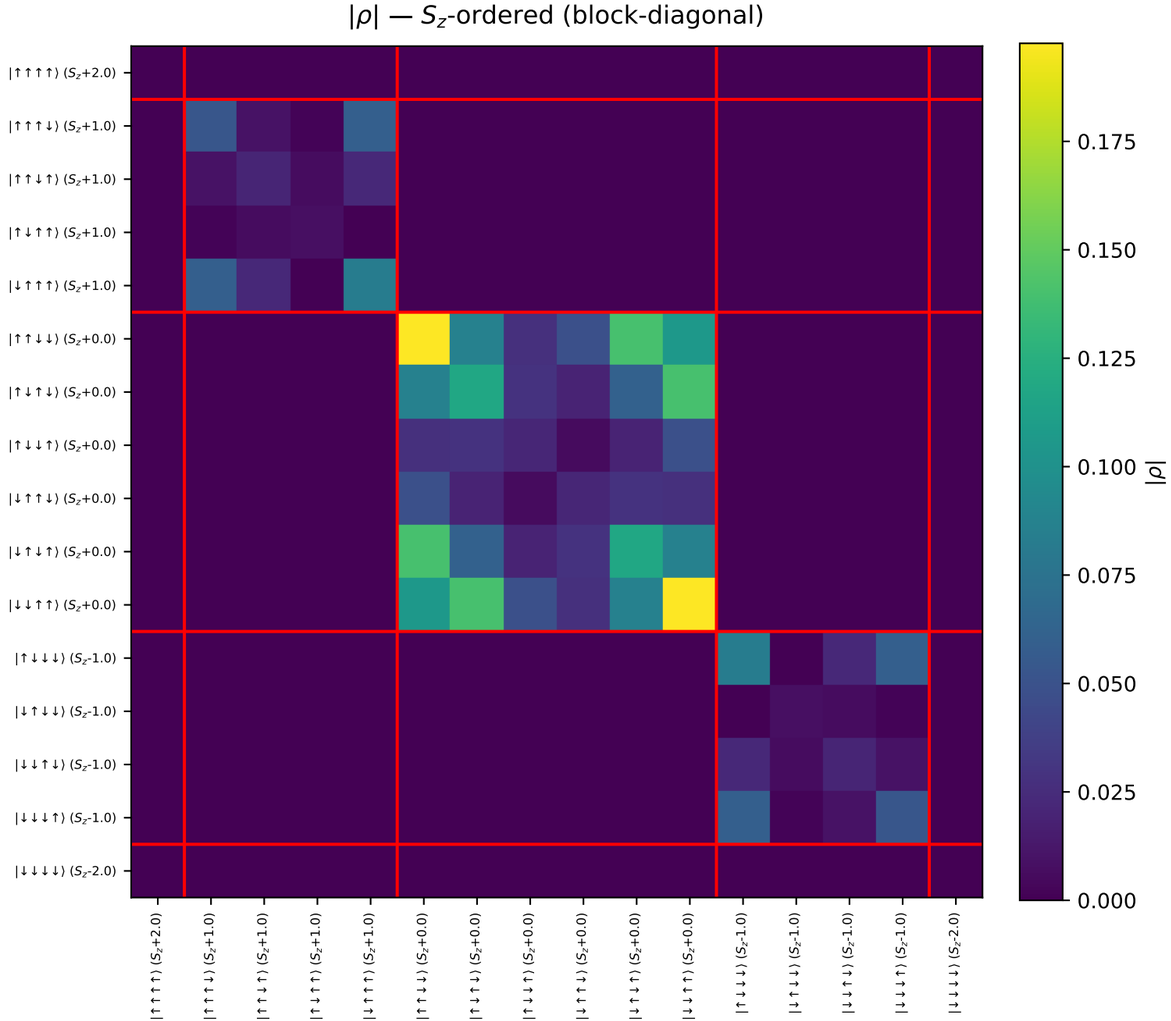


Trace=1.000000 Purity=0.158254 Entropy=2.042876 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.048435 ≤ C_{UB} = 0.411075 verdict: GME

4-Body RDM (Sites 1,2,3,5)

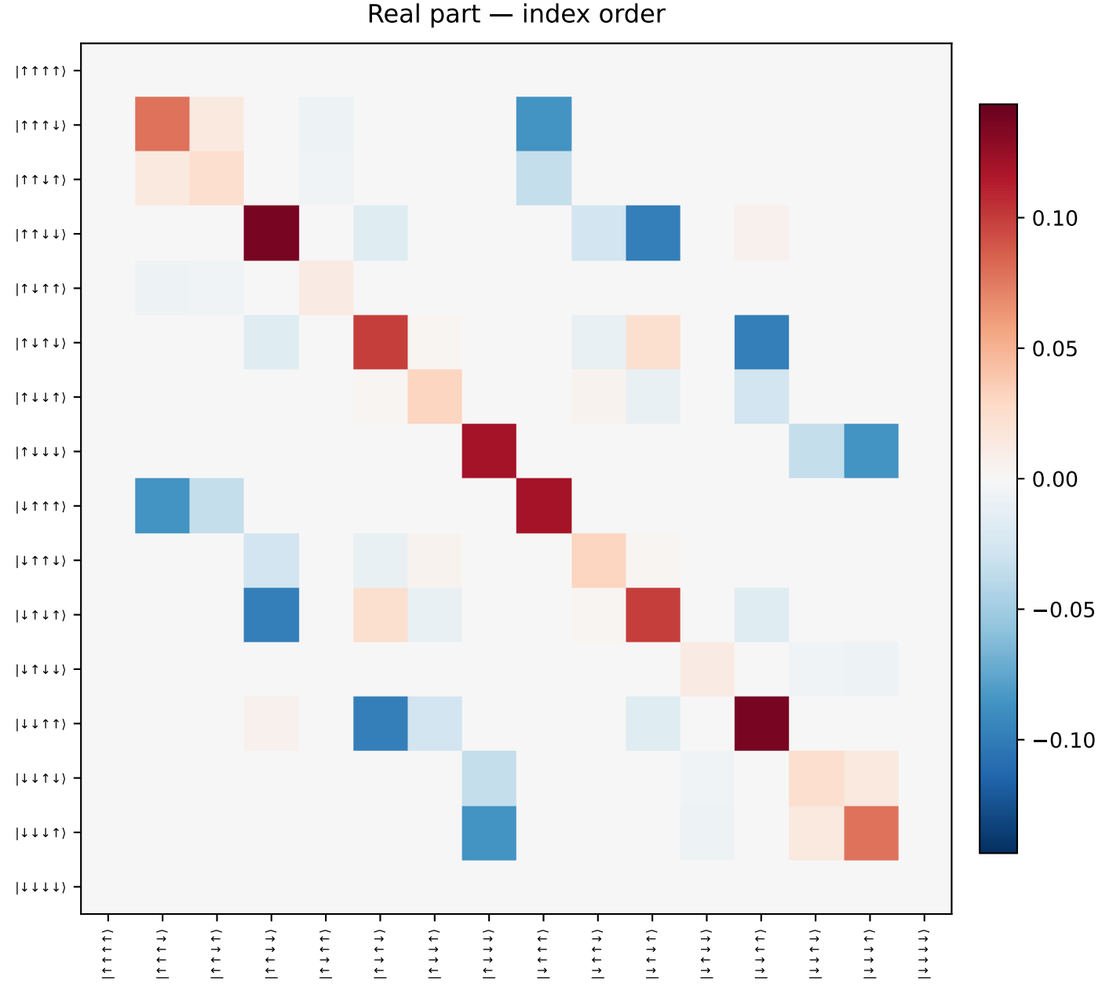


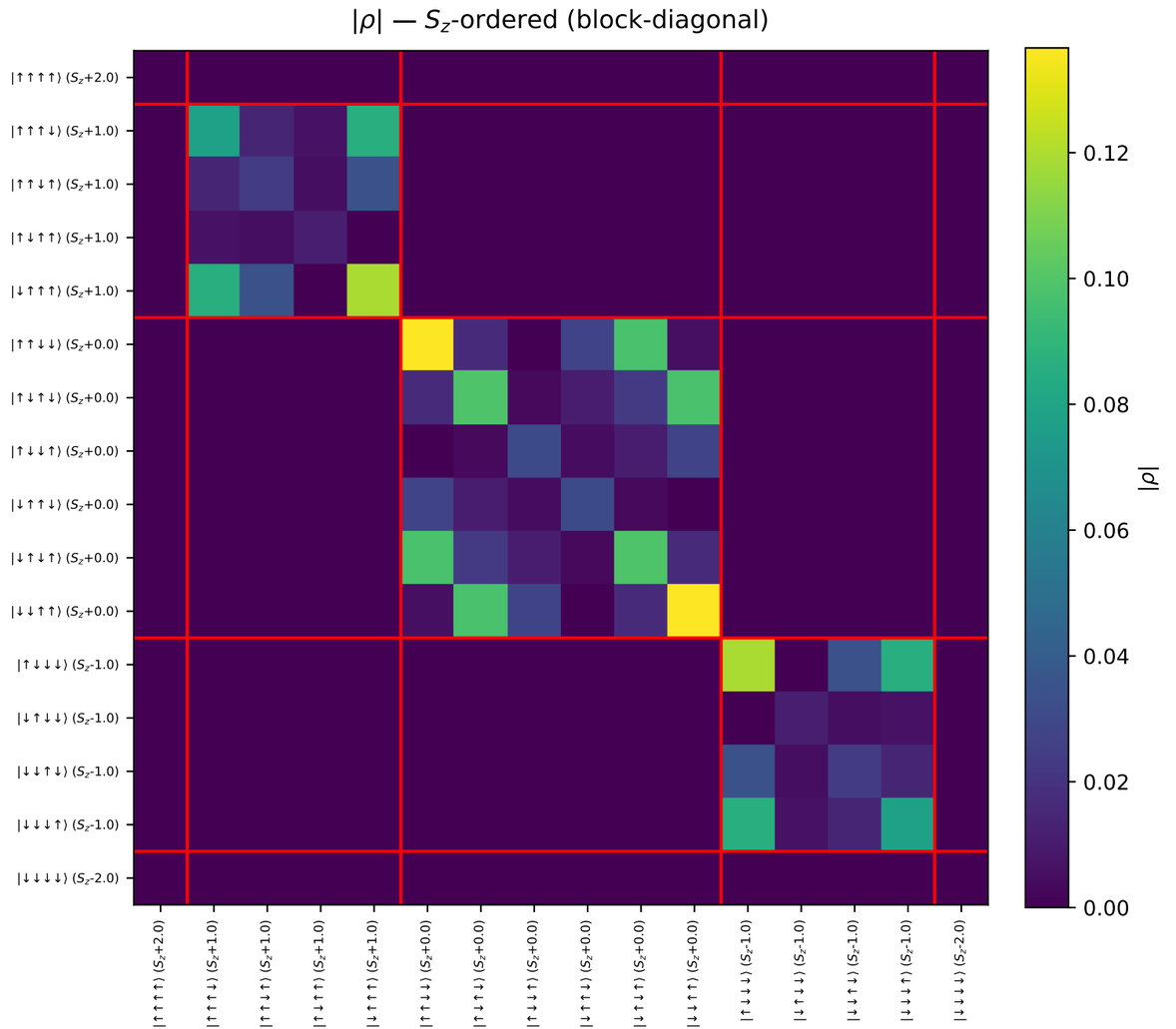


Trace=1.000000 Purity=0.300189 Entropy=1.582016 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.020941 ≤ C_{UB} = 0.344744 verdict: GME

4-Body RDM (Sites 1,2,4,5)

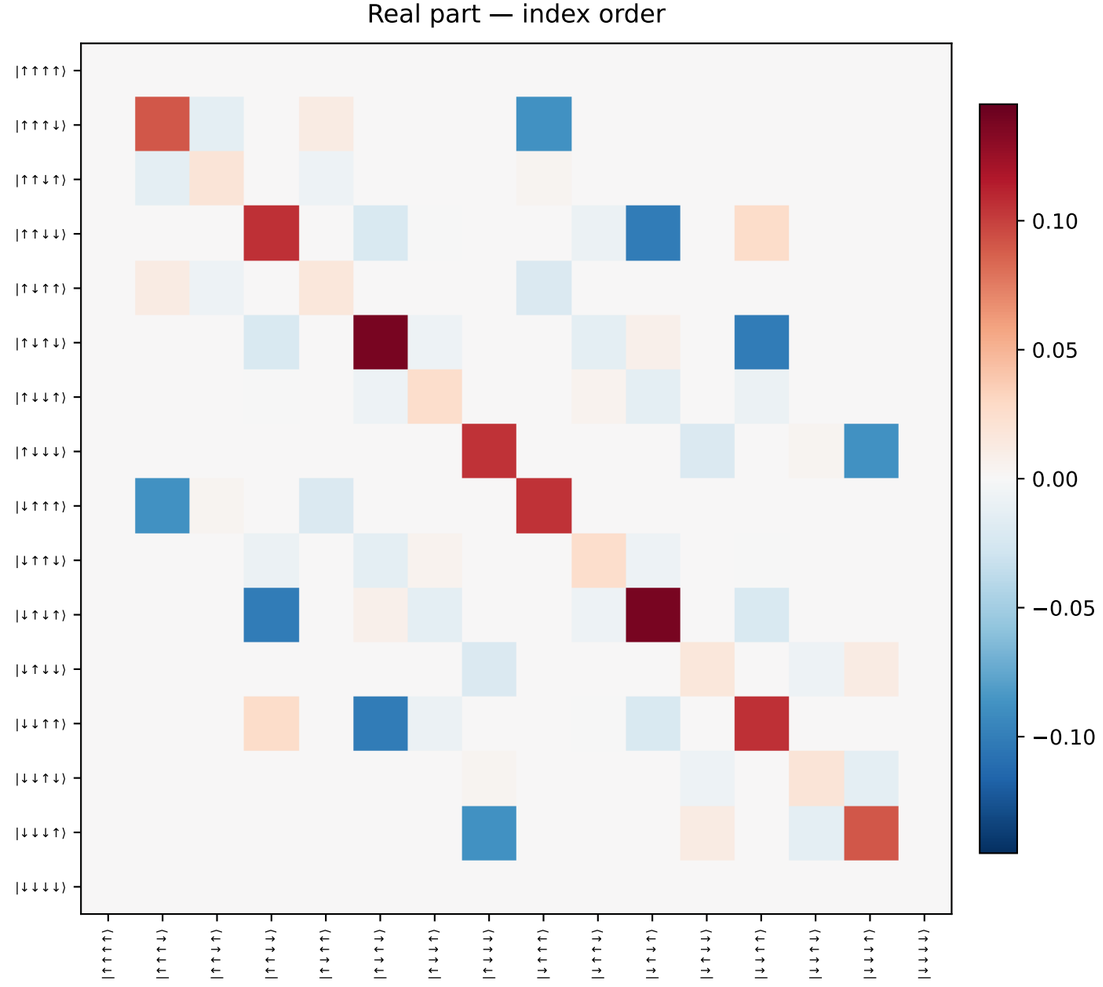


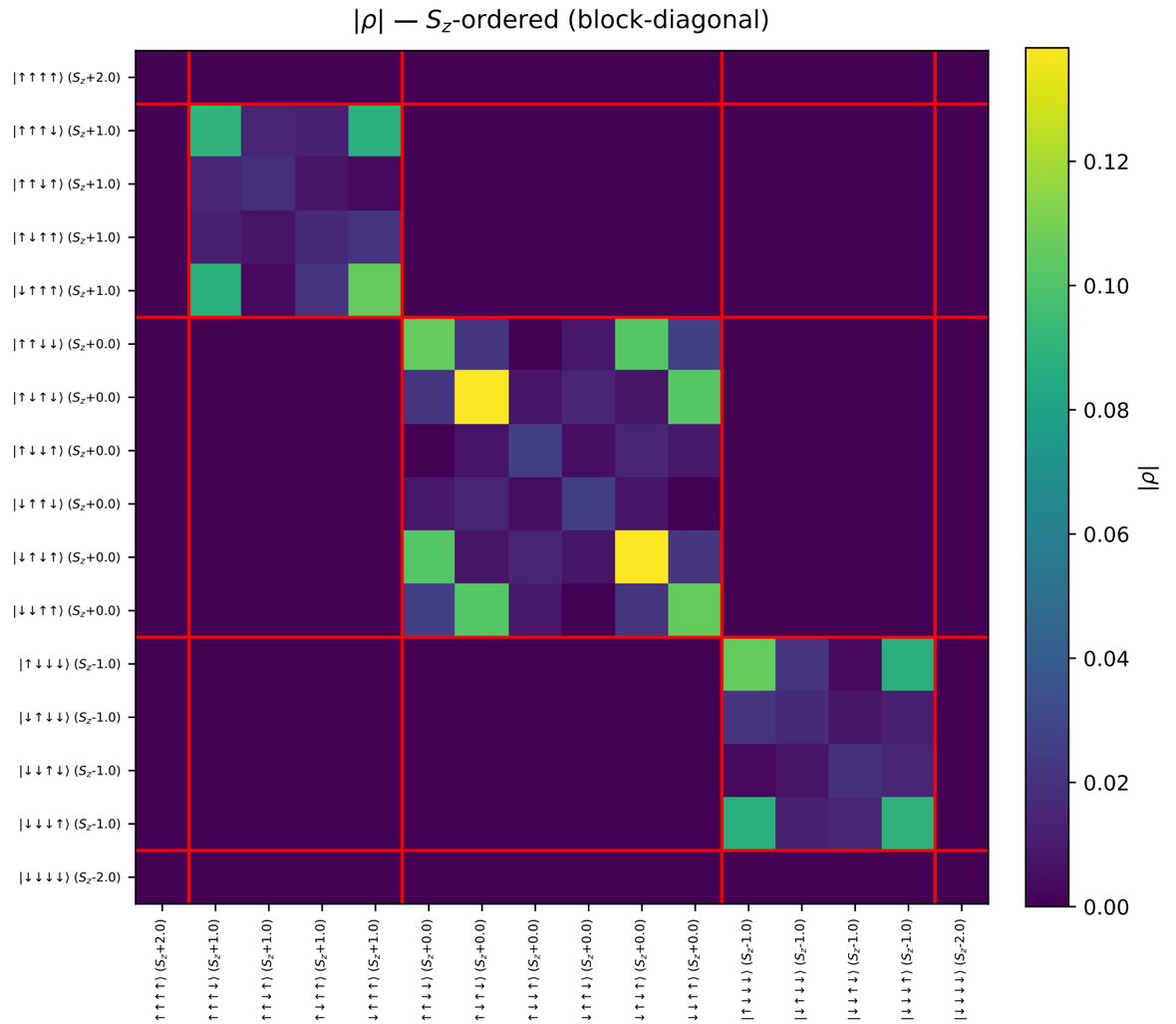


Trace=1.000000 Purity=0.179918 Entropy=1.912561 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.039429 ≤ C_{UB} = 0.242406 verdict: GME

4-Body RDM (Sites 1,3,4,5)

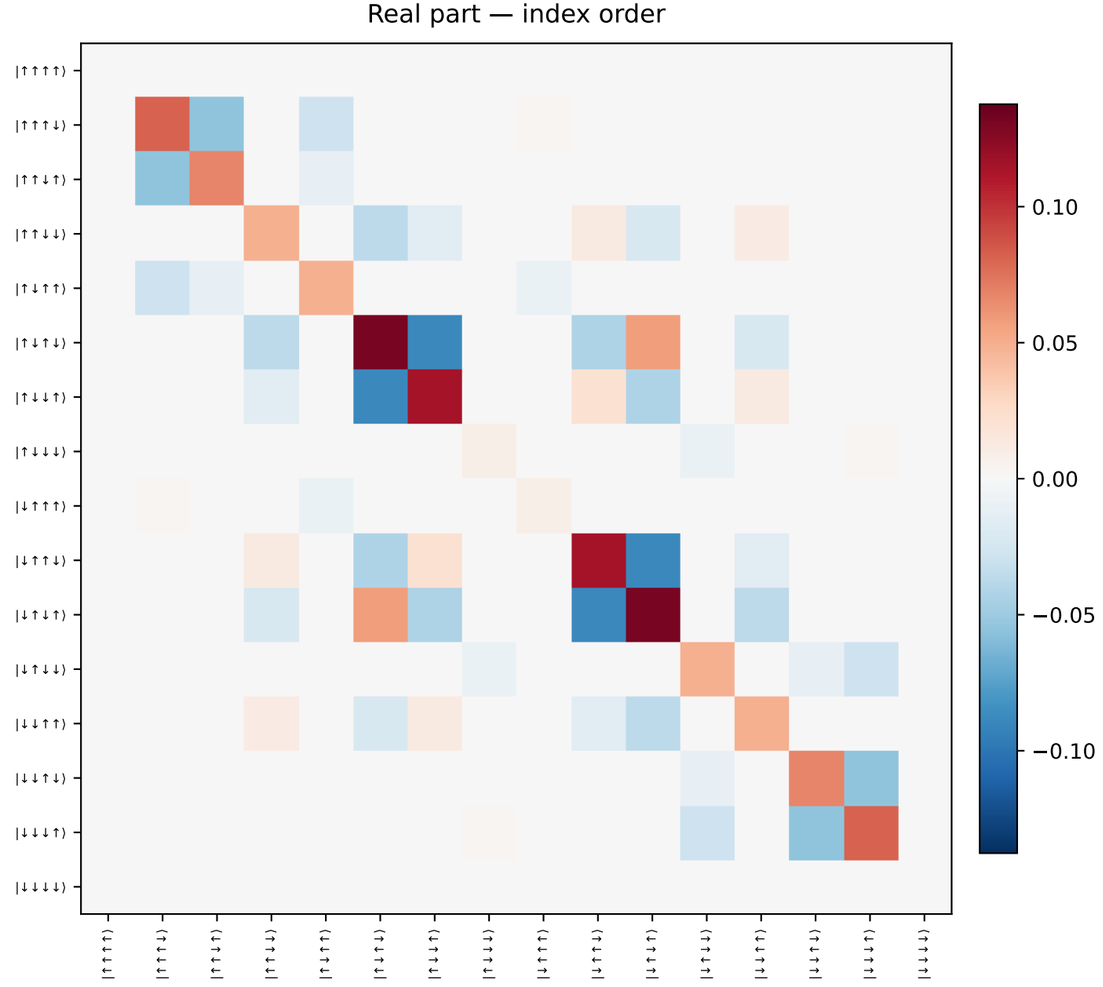


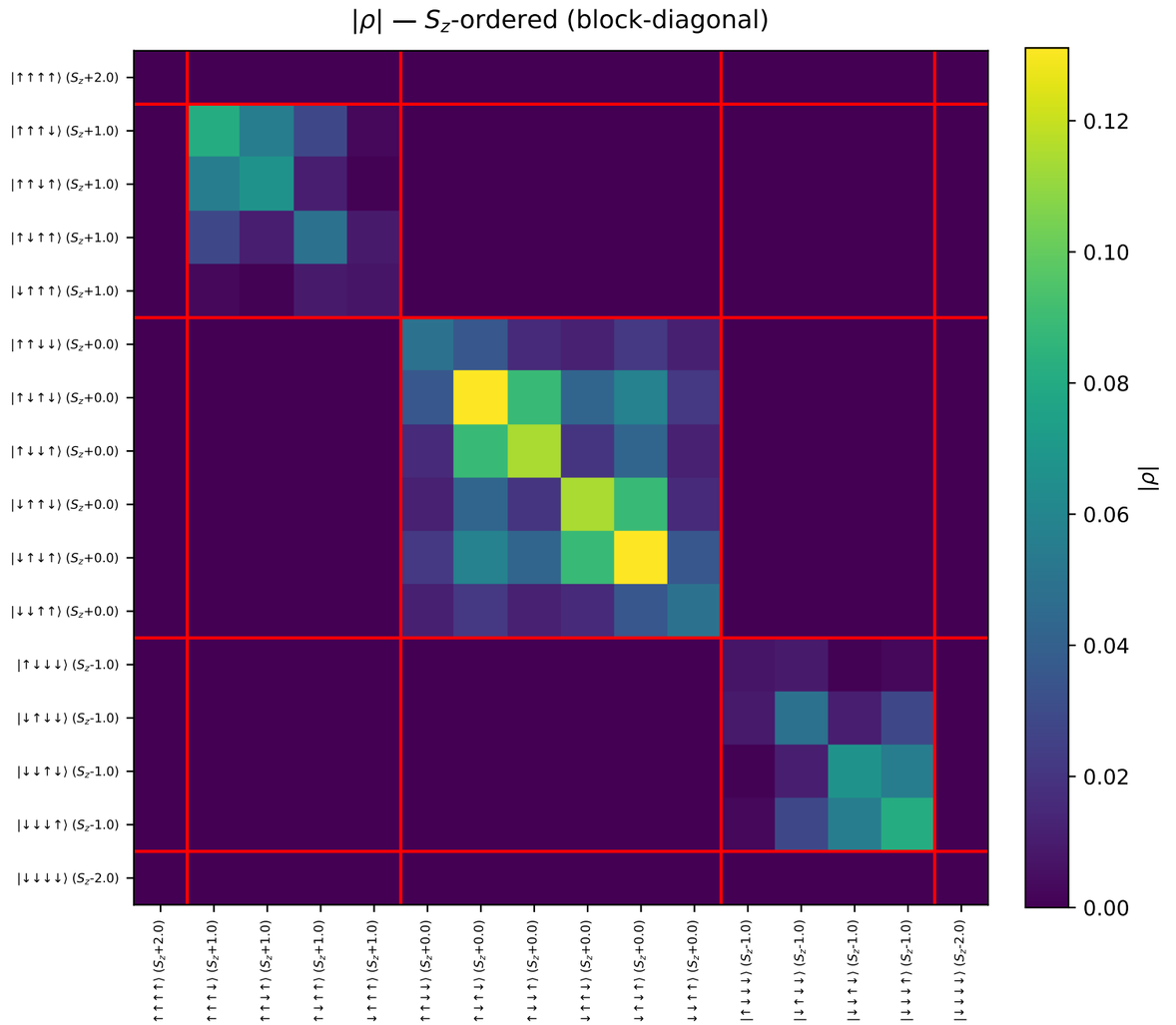


Trace=1.000000 Purity=0.181960 Entropy=1.911215 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.058349 ≤ C_{UB} = 0.245925 verdict: GME

4-Body RDM (Sites 2,3,4,5)





Trace=1.000000 Purity=0.164352 Entropy=2.016905 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.024065 ≤ C_{UB} = 0.338947 verdict: GME